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Spot beta estimation with asynchronous noisy prices

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We introduce a consistent estimator of the spot asset beta based on the Fourier estimator of the multivariate spot volatility. Firstly, we prove the rate-efficient pointwise asymptotic normality for the Fourier spot covariance estimator in the presence of asynchronous observations and microstructure noise. Using this result, we obtain a consistent estimator of the spot beta. Then, we investigate the performance of this estimator when employed to reconstruct the sample path of the spot asset beta. In this regard, we provide simulation results that suggest that our spot beta estimator has a robust finite-sample performance in the presence of realistic market features such as rough volatility, inhomogeneous asynchronous sampling, autocorrelated and price-dependent noise and price rounding. Additionally, we conduct an empirical study with tick-by-tick prices of 100 US stocks, sampled over the period January 1, 2023–December 31, 2023, in which we reconstruct intraday spot beta paths and obtain evidence of the existence of a statistically significant link between the intraday cross-sectional dispersion of spot betas and the intraday cross-sectional average and standard deviation of the microstructure noise variance, used as an inverse proxy of liquidity and informational efficiency. This empirical evidence suggests that intraday changes in microstructural features of the market might contribute to explaining high-frequency beta dynamics.

Keywords: Nonparametric spot covariance estimation; Beta coefficient; Epps effect; Fourier analysis; High-frequency data; Non-synchronicity

JEL Classification: C14, C32, G1

1. Introduction

The availability of reliable estimates of a stock's beta, namely, the ratio of the covariance between the market and the stock and the variance of the market, is of paramount importance for the accurate implementation of asset pricing models. While traditional approaches to beta estimation rely on linear regressions performed with low-frequency (e.g. daily or monthly) price series (see, e.g. Black *et al.* 1972), in recent years, the literature on financial econometrics has devoted particular attention to the use of high-frequency prices to improve non-parametric inference on betas, increasing the precision of estimates and showing that the assumption of a constant beta is rejected by empirical data even for short horizons, e.g. at the daily scale. However, while a number of works focused on reconstructing time-varying betas as the ratio of integrated

quantities (see, e.g. Kalnina 2023 and the references therein), fewer works have addressed the issue of making inferences on spot betas as a random function of time (see Bibinger *et al.* 2019, Andersen *et al.* 2021).

The efficient estimation of spot beta paths crucially depends on the accurate estimation of the spot covariance between the asset and the market, which can be particularly challenging at high frequencies. In the last twenty years, a number of papers (see Bollerslev and Zhang 2003, Barndorff-Nielsen and Shephard 2004, Hayashi and Yoshida 2005, Christensen *et al.* 2010, Griffin and Oomen 2011, among others), have shown the potential of the usage of high-frequency data for the non-parametric estimation of covariances and provided estimation methodologies to overcome the two main difficulties that arise when estimating covariances at high frequencies. The first one refers to the so-called Epps effect (see Epps 1979), that is, the fact that the asynchronicity of the arrival times of trades makes correlations among stocks biased

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toward zero as the sampling frequency increases.[†] The second one relates to the distortion of efficient prices caused by the presence of market microstructure contaminations. Specifically, estimators of integrated covariance that are robust to both asynchronicity and microstructure noise were proposed by Aït-Sahalia *et al.* (2010), Barndorff-Nielsen *et al.* (2011), Zhang (2011), Bibinger *et al.* (2014) and Koike (2014).

Unlike the case of integrated covariance, the non-parametric high-frequency estimation of the instantaneous (or spot) covariance represents a more recent and less-investigated topic, see Malliavin and Mancino (2002), Malliavin and Mancino (2009), Park *et al.* (2016) and Bibinger *et al.* (2019). Specifically, Malliavin and Mancino (2002) first proposed a Fourier analysis-based method for multivariate spot volatility estimation and proved its uniform consistency in the presence of asynchronous observations. Later, Park *et al.* (2016) proposed a multivariate spot volatility estimator that is robust both to the microstructure noise and asynchronous sampling. This estimator is based on Fourier-domain techniques, although it employs a more general kernel weighting, and has the advantage that it does not require an explicit time alignment of observations. Park *et al.* (2016) showed that their approach provides a uniformly consistent estimator of the instantaneous co-volatility, even in the presence of quite general dependent microstructure noise. Further, the authors provided a central limit theorem for the estimator of any Fourier coefficient of the series expansion. More recently, Bibinger *et al.* (2019) introduced a spot covariance estimator based on a local method of moments (LMM) and showed its consistency and the asymptotic normality of the related estimation error in the presence of noise and asynchronous sampling.

In this paper, we prove the rate-efficient pointwise asymptotic normality for the Fourier spot covariance estimator in the presence of asynchronous observations and microstructure noise. This result substantially extends (Malliavin and Mancino 2009), where the weak convergence for the Fourier spot covariance estimator was obtained with asynchronous observations but only in the absence of microstructure noise. Moreover, it complements the one in Park *et al.* (2016) by proving the rate of convergence of the Fourier spot covariance estimator. Thanks to the pointwise convergence result, we then obtain a consistent estimator of the spot beta.

Furthermore, we study the finite-sample robustness of the spot beta estimator in an extensive numerical exercise. After introducing a feasible method for the selection of the cutting frequencies involved in the implementation of the spot beta estimator, we evaluate its performance in different scenarios characterized by progressively more realistic features of the microstructure noise and the sampling scheme. These include non-homogeneous Poisson sampling rates, autocorrelated and time-dependent noise processes and price rounding. Moreover, we consider not only the case when the volatility of the efficient price is driven by a Brownian motion but also

the case of a rough volatility model. The numerical results we obtain suggest that our spot beta estimator provides satisfactory efficiency in all scenarios considered, thereby supporting its robustness for empirical applications.

As mentioned, the performance of the spot beta estimator crucially depends on the efficiency of the spot covariance estimator in dealing with asynchronicity and noise issues. Accordingly, we also provide numerical results supporting the robustness of the Fourier spot covariance estimator at high frequencies under different levels of the relative liquidity of the equity asset with respect to the market. Additionally, we compare the finite-sample performance of the proposed spot beta estimator to that of the LMM spot beta estimator by Bibinger *et al.* (2019) and the realized spot beta estimator used in Andersen *et al.* (2021). As a result, in all the scenarios considered we obtain numerical evidence suggesting that our estimator outperforms both the LMM and realized estimators.

Finally, we employ the proposed spot beta estimator for an empirical application, in which we test the existence of a statistically significant relationship between the intraday cross-sectional dispersion of spot betas and the intraday cross-sectional mean and standard deviation of the (estimated) microstructure noise variance. Based on the empirical results by Andersen *et al.* (2021), spot betas are typically not constant throughout the trading day and show time-varying patterns. We argue that such patterns may be linked to those displayed by the microstructure noise variance for economic reasons. For instance, a decrease in the microstructure noise variance could be associated with larger liquidity and lower transaction costs (see, e.g. Aït-Sahalia and Yu 2009), which in turn could reduce the perceived risk and the desired expected return, that is, the asset's beta. In the case of beta dispersion, the latter may be positively dependent on the cross-sectional mean and standard deviation of the microstructure noise variance: for example, near market open and market close, when microstructure noise variance tends to be larger and more volatile across assets, more pronounced idiosyncratic differences in liquidity and informational efficiency may generate a larger dispersion in betas.

We empirically test the existence of an affine link between the logarithm of the cross-sectional dispersion of spot betas and, respectively, the cross-sectional mean and standard deviation of the noise variance, using tick-by-tick prices of 100 US stocks sampled over the period January 1, 2023–December 31, 2023. As a result, we find statistically significant results both in the case when the cross-sectional average and standard deviation of the noise variance are used as regressors. However, the cross-sectional standard deviation of the noise provides a more satisfactory fit based on the comparison of R^2 , AIC and BIC values.

The paper is organized as follows. Section 2 introduces the setting and the assumptions behind our study. Section 3 recalls the definition of the Fourier spot covariance estimator and proves asymptotic results. Sections 4 and 5 present, respectively, the numerical study and the empirical exercise. Finally, section 6 concludes. The appendices contain the proofs, auxiliary results on the Dirichlet and Fejér kernels, additional numerical results and the list of stocks employed in the numerical study.

[†] In a low-frequency setting, the misspecification of beta estimates in the presence of asynchronous sampling and the ensuing covariance bias has already been pointed out by the early contribution of Scholes and Williams (1977) and, more recently, by Frazzini and Pedersen (2014).

2. Setting and assumptions

In this section, we introduce the data-generating model and present the Fourier method for multivariate spot volatility estimation, which will be employed for estimating the spot beta. We will assume that the efficient prices and the latent volatilities and covariances are continuous semimartingales, but observed prices are contaminated by microstructure noise.

Specifically, the following assumptions on the efficient price process, the sampling scheme and the noise process are made.

ASSUMPTION (P) The log-price process p satisfies the following assumptions.

(P.I) The log-price process p is assumed to be a d -dimensional continuous Itô semimartingale satisfying

$$dp^j(t) = \sum_{r=1}^K \sigma_{j,r}(t) dW_t^r + b_j(t) dt, \quad j = 1, \dots, d,$$

where $W = (W^1, \dots, W^K)$ is a K -dimensional Brownian motion on a filtered probability space $(\Omega, (\mathcal{F}_t)_{t \in [0, T]}, P)$, which satisfies the usual conditions.

The spot covariance process is defined, for any $1 \leq j, j' \leq d$, as

$$\Sigma^{jj'}(t) = \sum_{r=1}^K \sigma_{j,r}(t) \sigma_{j',r}^*(t) = \sigma_j^*(t) \sigma_{j'}(t),$$

where σ_j^* denotes the transpose of the vector σ_j . Moreover, we assume $\Sigma^{jj'}(t)$ is an Itô semimartingale, and $\gamma_{jj'}(t)$ will denote the diffusion coefficient of the covariance process $\Sigma^{jj'}(t)$.[†]

(P.II) For any $j, j' = 1, \dots, d$ and $r = 1, \dots, K$, $\sigma_{j,r}$, b_j and $\gamma_{jj'}$ are continuous and adapted stochastic processes defined on the same probability space $(\Omega, (\mathcal{F}_t)_{t \in [0, T]}, P)$ and such that, for any $q \geq 1$:

$$\begin{aligned} E \left[\int_0^T \sigma_{j,r}^q(t) dt \right] &< \infty, & E \left[\int_0^T b_j^q(t) dt \right] &< \infty, \\ E \left[\int_0^T \gamma_{jj'}^q(t) dt \right] &< \infty. \end{aligned}$$

The processes are specified in such a way that $\sigma_{j,r}$ and $\gamma_{jj'}$ are a.s. positive.

Let $0 = t_0^j < \dots < t_{n_j}^j = T$ denote the observation times for the j th asset. The sampling scheme obeys the following assumption.

ASSUMPTION (S) Let

$$\rho(n_j) = \sup_{0 \leq h \leq n_j-1} |t_{h+1}^j - t_h^j|, \quad \rho(n) = \sup_{j=1, \dots, d} \rho(n_j)$$

and

$$\begin{aligned} H_n^{jj'}(t) &= (n_j \wedge n_{j'}) \sum_{t_{h+1}^j \wedge t_{k+1}^{j'} < t} (t_{h+1}^j \wedge t_{k+1}^{j'} - t_h^j \vee t_k^{j'}) I_{\{t_h^j \vee t_k^{j'} < t_{h+1}^j \wedge t_{k+1}^{j'}\}}. \end{aligned}$$

We assume that:

(S.I) $\rho(n_j) = s_j \rho(n) + o(\rho(n))$, with s_j denoting a positive constant, for $j = 1, \dots, d$;

(S.II) $\sup_{0 \leq h \leq n_j-1, 0 \leq k \leq n_{j'}-1} |t_h^j - t_k^{j'}| I_{\{[t_{h-1}^j - t_h^j] \cap [t_{k-1}^{j'} - t_k^{j'}] \neq \emptyset\}} = O((n_j \wedge n_{j'})^{-1})$;

(S.III) $\lim_{n \rightarrow \infty} H_n^{jj'}(t) = H^{jj'}(t)$, and suppose that $H^{jj'}(t)$ is continuously differentiable.

We are now in the position to introduce the definition of the Fourier estimator of spot covariance, according to Malliavin and Mancino (2009). For $j = 1, \dots, d$, define the discrete Fourier coefficients of the log-returns as

$$c_k(dp_{n_j}^j) := \frac{1}{T} \sum_{i=0}^{n_j-1} e^{-it_i^j \frac{2\pi}{T} k} \delta_i(p^j),$$

with $\delta_i(p^j) := p(t_{i+1}^j) - p(t_i^j)$, and, for any j, j' , define an unbiased estimator of the k th Fourier coefficient of the covariance process, through the following convolution formula[‡]:

$$c_k(\Sigma_{n,N}^{jj'}) := \frac{T}{2N+1} \sum_{|h| \leq N} c_h(dp_{n_j}^j) c_{k-h}(dp_{n_{j'}}^{j'}). \quad (1)$$

Finally, for any $t \in (0, T)$, the spot covariance estimator is defined as

$$\widehat{\Sigma}_{n,N,M}^{jj'}(t) := \sum_{|k| \leq M} \left(1 - \frac{|k|}{M+1}\right) e^{it \frac{2\pi}{T} k} c_k(\Sigma_{n,N}^{jj'}). \quad (2)$$

By means of the rescaled Dirichlet kernel $D_N(x) := \frac{1}{2N+1} \sum_{|k| \leq N} e^{ik \frac{2\pi}{T} x}$, it is possible to express (1) as

$$c_k(\Sigma_{n,N}^{jj'}) = \frac{1}{T} \sum_{i=0}^{n_j-1} \sum_{i'=0}^{n_{j'}-1} D_N(t_i^j - t_{i'}^{j'}) e^{-ik t_i^j \frac{2\pi}{T}} \delta_i(p^j) \delta_{i'}(p^{j'}).$$

Therefore, the Fourier spot volatility estimator (2) can be rewritten, using two kernels, as

$$\widehat{\Sigma}_{n,N,M}^{jj'}(t) = \frac{1}{T} \sum_{i=0}^{n_j-1} \sum_{i'=0}^{n_{j'}-1} F_M(t - t_{i'}^{j'}) D_N(t_i^j - t_{i'}^{j'}) \delta_i(p^j) \delta_{i'}(p^{j'}),$$

where $F_M(x) := \sum_{|k| \leq M} (1 - \frac{|k|}{M+1}) e^{ik \frac{2\pi}{T} x}$ is the Fejér kernel.

[†] The Fourier method for spot covariance estimation does not require the volatility process to be a Brownian semimartingale, as it only assumes the volatility process to be a random function with Hölder-continuous paths. Under this condition, the Fourier estimator achieves a slightly sub-optimal rate of convergence, see Mancino and Recchioni (2015). Therefore, rough volatility models are also included at the cost of a slower convergence rate.

[‡] Hereafter, we accept abuse of notation by denoting the estimator $\widehat{\Sigma}_{n_1, n_2, N, M}^{jj'}(t)$ as $\widehat{\Sigma}_{n, N, M}^{jj'}(t)$.

Finally, define the observed log-price process, contaminated by microstructure noise, as

$$\tilde{p}_{t_h}^j := p_{t_h}^j + \eta_{t_h}^j, \quad j = 1, \dots, d, \quad h = 0, \dots, n_j,$$

where p denotes the efficient log-price process while η represents the additive noise due to the presence of market microstructure, which is specified as follows.

ASSUMPTION (N) The additive noise term η is such that:

- (N.I) $E[\eta_{t_h}^j] = 0$ and $E[(\eta_{t_h}^j)^2] = \xi_j, \forall h, j$;
- (N.II) $\eta_{t_h}^j$ and $\eta_{t_{h'}}^{j'}$ are i.i.d. for $h \neq h'$;
- (N.III) $\eta_{t_h}^j$ is independent of $\eta_{t_{h'}}^{j'}$ for $j \neq j'$ and any h, h' ;
- (N.IV) η is independent of p .

The Fourier spot covariance estimator in the presence of additive noise reads as

$$\tilde{\Sigma}_{n,N,M}^{jj'}(t) := \sum_{|k| \leq M} \left(1 - \frac{|k|}{M+1}\right) e^{it \frac{2\pi}{T} k} c_k(\tilde{\Sigma}_{n,N}^{jj'}), \quad (3)$$

where

$$c_k(\tilde{\Sigma}_{n,N}^{jj'}) := \frac{T}{2N+1} \sum_{|h| \leq N} c_h(d\tilde{p}_{n_j}^j) c_{k-h}(d\tilde{p}_{n_{j'}}^{j'}) \quad (4)$$

and

$$c_k(d\tilde{p}_{n_j}^j) := \frac{1}{T} \sum_{i=0}^{n_j-1} e^{-it_i^j \frac{2\pi}{T} k} \delta_i(\tilde{p}^j). \quad (5)$$

We note that, in the noisy case, the Fourier spot covariance estimator maintains the same form as (2), but uses the discrete Fourier coefficients of the noisy log-price in (5).

3. Asymptotic results

The main result of this section is obtaining a consistent estimator of spot beta, in the presence of microstructure noise and asynchronicity. To this aim, we first prove that the estimator in (3) is consistent and rate-efficient in the presence of noise and asynchronous sampling, without the need for any bias correction or preliminary price manipulation. The next Theorem 3.1 proves the asymptotic normality for the estimator (3). For simplicity, in the following, we assume that $j = 1$ and $j' = 2$. A simulation study that offers support for the normality of the standardized estimation error from Theorem 3.1 with finite samples is detailed in Appendix C.

THEOREM 3.1 *Let Assumptions (P), (S) and (N) hold. Moreover, $N\rho(n)^{1/2} \sim c > 0$ and $MN^{-1/2} \sim a > 0$. Then, as $\rho(n) \rightarrow 0, N, M \rightarrow \infty$, the following stable convergence in*

law holds, for any fixed $t \in (0, T)$:

$$\begin{aligned} & \rho(n)^{-1/4} M^{-1/2} \left(\tilde{\Sigma}_{n,N,M}^{1,2}(t) - \Sigma^{1,2}(t) \right) \\ & \downarrow \\ & \mathcal{N} \left(0, \frac{1}{3c} (\Sigma^{1,1}(t) \Sigma^{2,2}(t) + (\Sigma^{1,2}(t))^2) \right. \\ & \quad + \frac{T}{3a^2c} \gamma_{1,2}^2(t) + \frac{c}{9} (\Sigma^{1,1}(t) s_2 \xi_2 + \Sigma^{2,2}(t) s_1 \xi_1) \\ & \quad \left. + \frac{c^3}{15} s_1 s_2 \xi_1 \xi_2 \right). \end{aligned}$$

REMARK 3.2 The rate of convergence is the optimal one for spot covariance estimation in the presence of noise, namely $n^{-1/8}$. This rate is the same as in Bibinger *et al.* (2019) and is the square root of the corresponding optimal rate for integrated covariance estimation in a setting with asynchronicity and noise, that is, $n^{-1/4}$ (see also Bibinger *et al.* 2019, Chen *et al.* 2024 on the rate optimality). It is worth noticing that the effects of asynchronicity and microstructure contamination are overcome at the same time as an effect of choosing $N\rho(n) \rightarrow 0$, that is, of cutting out the highest frequencies N . A finite-sample study and an extensive simulated analysis showing that $N \ll n$ is the correct choice in the presence of microstructure noise can be found in Mancino and Sanfelici (2011). In the absence of noise, the number of Fourier frequencies N can be chosen of the same order as n , and, in particular, in the univariate case, the Nyquist frequency $N = n/2$ represents the optimal choice in terms of asymptotic-variance minimization, see Mancino and Recchioni (2015). However, in the multivariate case, even in the absence of microstructure noise, under such choice a bias due to asynchronous sampling would appear and the asymptotic variance would depend on the quadratic variation of time $H'(t)$, see also Clement and Gloter (2011). Therefore, the efficient choice of N in the presence of asynchronicity and microstructure noise is $N\rho(n) \rightarrow 0$.

The asymptotic normality for the Fourier estimator of the spot variance $\tilde{\Sigma}^{2,2}(t)$ is obtained in Mancino *et al.* (2024). Therefore, we can obtain a consistent estimator of the spot beta, defined as

$$\beta^{1,2}(t) := \frac{\Sigma^{1,2}(t)}{\Sigma^{2,2}(t)}. \quad (6)$$

In virtue of the Theorem 3.1, in the presence of noise and asynchronicity, the Fourier spot beta estimator reads as

$$\tilde{\beta}_{n,N,M}^{1,2}(t) := \frac{\tilde{\Sigma}_{n,N,M}^{1,2}(t)}{\tilde{\Sigma}_{n,N,M}^{2,2}(t)}, \quad (7)$$

where $\tilde{\Sigma}_{n,N,M}^{jj'}(t), j, j' \in \{1, 2\}$, is defined in (3). The following result provides the consistency of the proposed estimator of spot beta with noisy observations.

PROPOSITION 3.3 *Let Assumptions (P), (S) and (N) hold. Moreover, $N\rho(n)^{1/2} \sim c > 0$, $MN^{-1/2} \sim a > 0$. Then, as $\rho(n) \rightarrow 0, N, M \rightarrow \infty$, the following convergence in probability holds, for any fixed $t \in (0, T)$:*

$$\tilde{\beta}_{n,N,M}^{1,2}(t) \rightarrow \tilde{\beta}^{1,2}(t). \quad (8)$$

4. Finite-sample results

In this section, we study the finite-sample performance of the spot beta estimator defined in (7) using simulations. Firstly, exploiting the asymptotic theory from Theorem 3.1, we introduce a feasible method for selecting the cutting frequencies N and M and evaluate its accuracy with respect to the unfeasible optimal selection. Then, we evaluate the robustness of this feasible method to different realistic features, namely, non-homogeneous sampling, autocorrelated and price-dependent noise, rounding and rough volatility. Finally, we provide a comparison of the finite-sample performance of the estimator with those of the realized spot beta estimator, used, e.g. in Andersen *et al.* (2021), and the LMM spot beta estimator by Bibinger *et al.* (2019). In particular, we focus on the impact of liquidity on the estimation of the spot covariance, that is, the numerator of the spot beta.

4.1. Feasible selection of the cutting frequencies

In this section, we illustrate a feasible method for selecting the cutting frequencies N and M and assess its efficiency with simulated prices. The method, which is inspired by the data-driven adaptive criterion developed by Li *et al.* (2022, Sect. 2.4), aims at minimizing the L^1 -norm of the vector of the integrated asymptotic variances of $\tilde{\Sigma}_{n,N,M}^{1,2}(t)$ and $\tilde{\Sigma}_{n,N,M}^{2,2}(t)$ in (7) on the interval $[0, T]$.[†]

4.1.1. Outline of the method. Based on Theorem 3.1, the integrated asymptotic variance of $\Sigma_{n,N,M}^{1,2}(t)$ on $[0, T]$, denoted by $\Phi(N, M)$, reads as

$$\begin{aligned}\Phi(N, M) &= \frac{M}{N} \frac{1}{3} \int_0^T q^{1,2}(t) dt + \frac{1}{M} \frac{T}{3} \int_0^T \gamma_{1,2}^2(t) dt \\ &\quad + MN \frac{1}{9} \int_0^T (\rho(n_2) \xi_2 \Sigma^{1,1}(t) + \rho(n_1) \xi_1 \Sigma^{2,2}(t)) dt \\ &\quad + MN^3 \rho(n_1) \rho(n_2) \frac{T}{15} \xi_1 \xi_2,\end{aligned}\quad (9)$$

where $q^{1,2}(t) := \Sigma^{1,1}(t) \Sigma^{2,2}(t) + (\Sigma^{1,2}(t))^2$. Similarly, as shown in Mancino *et al.* (2024, Theorem 3.6), the integrated asymptotic variance of $\Sigma_{n,N,M}^{2,2}(t)$ on $[0, T]$, which we denote by $\Psi(N, M)$, is given by

$$\begin{aligned}\Psi(N, M) &= \frac{M}{N} \frac{1}{3} \int_0^T q^{2,2}(t) dt + \frac{1}{M} \frac{T}{3} \int_0^T \gamma_{2,2}^2(t) dt \\ &\quad + MN \rho(n_2) \frac{4}{9} \xi_2 \int_0^T \Sigma^{2,2}(t) dt \\ &\quad + MN^3 \rho(n_2)^2 \frac{T}{15} \xi_2^2.\end{aligned}\quad (10)$$

The Fourier method allows reconstructing (9) and (10) with high-frequency log-prices. Specifically, the estimators of (9)

and (10) read as

$$\begin{aligned}\hat{\Phi}(N, M) &= \frac{M}{N} \frac{T}{3} c_0(\tilde{q}_{n,N,S}^{1,2}) + \frac{1}{M} \frac{T^2}{3} c_0((\tilde{\gamma}_{1,2})_{n,N,S}^2) \\ &\quad + MN \frac{T}{9} \left(\rho(n_2) \hat{\xi}_{n_2}^2 c_0(\tilde{\Sigma}_{n,N}^{1,1}) + \rho(n_1) \hat{\xi}_{n_1}^2 c_0(\tilde{\Sigma}_{n,N}^{2,2}) \right) \\ &\quad + MN^3 \rho(n_1) \rho(n_2) \frac{T}{15} \hat{\xi}_{n_1}^2 \hat{\xi}_{n_2}^2\end{aligned}\quad (11)$$

and

$$\begin{aligned}\hat{\Psi}(N, M) &= \frac{M}{N} \frac{T}{3} c_0(\tilde{q}_{n,N,S}^{2,2}) + \frac{1}{M} \frac{T^2}{3} c_0((\tilde{\gamma}_{2,2})_{n,N,S}^2) \\ &\quad + MN \rho(n_2) \frac{4}{9} \hat{\xi}_{n_2}^2 T c_0(\tilde{\Sigma}_{n,N}^{2,2}) \\ &\quad + MN^3 \rho(n_2)^2 \frac{T}{15} (\hat{\xi}_{n_2}^2)^2,\end{aligned}\quad (12)$$

where: for any $1 \leq j \leq 2$, $c_0(\tilde{\Sigma}_{n,N}^{jj})$ can be obtained from (4) by setting $k = 0$; for any $1 \leq j, j' \leq 2$, $c_0(\tilde{q}_{n,N,S}^{jj'})$ and $c_0((\tilde{\gamma}_{jj'})_{n,N,S}^2)$ are defined, for $S \sim \sqrt{N}$, respectively, as[‡]

$$\begin{aligned}c_0(\tilde{q}_{n,N,S}^{jj'}) &:= \sum_{|h| \leq S} c_h(\tilde{\Sigma}_{n,N}^{jj}) c_{-h}(\tilde{\Sigma}_{n,N}^{j'j'}) \\ &\quad + \sum_{|h| \leq S} c_h(\tilde{\Sigma}_{n,N}^{jj'}) c_{-h}(\tilde{\Sigma}_{n,N}^{jj})\end{aligned}\quad (13)$$

and

$$\begin{aligned}c_0((\tilde{\gamma}_{jj'})_{n,N,S}^2) &:= \frac{T}{S+1} \sum_{|h| \leq S} \left(1 - \frac{|h|}{S+1}\right) h^2 c_h(\tilde{\Sigma}_{n,N}^{jj'}) c_{-h}(\tilde{\Sigma}_{n,N}^{jj'})\end{aligned}\quad (14)$$

(for further details, see Mancino *et al.* (2017, Chapters 3 and 5), and the references therein); for any $1 \leq j \leq 2$, following Bandi and Russell (2006), $\hat{\xi}_{n_j}^j$ is defined as

$$\hat{\xi}_{n_j}^j := \frac{1}{2n_j} \sum_{i=0}^{n_j-1} \delta_i(p^j)^2.\quad (15)$$

For given sample sizes n_1 and n_2 , we aim at finding the values of N and M that minimize the L^1 -norm of the vector $(\hat{\Phi}(N, M), \hat{\Psi}(N, M))$, which we denote by $\Xi(N, M) := \hat{\Phi}(N, M) + \hat{\Psi}(N, M)$. Specifically, the optimization problem can be formulated as

$$\min_{(N, M) \in R} \Xi(N, M),$$

where $R = [\underline{N}, \bar{N}] \times [\underline{M}, \bar{M}] \subset \mathbb{R}^+ \times \mathbb{R}^+$ is such that $M < N < \min(n_1, n_2)$. To this aim, we employ the following method, based on the gradient-descent algorithm.

[†] The integrated asymptotic variance of the estimation error is also employed as the objective function for the optimal selection of the tuning parameters of spot volatility estimators by Zu and Boswijk (2014) and Figueroa-Lopez and Wu (2022).

[‡] Recall that $\gamma_{jj'}(t)$ is the diffusion coefficient of the variance function $\Sigma^{jj'}(t)$. Unreported simulations suggest that a suitable choice for S is given by $S = \lfloor 0.1\sqrt{N} \rfloor$.

Adaptive method for the selection of (N, M)

- Given the initial condition $(N_0 = \underline{N}, M_0 = \underline{M})$, for $k = 1, \dots, K$, follow the update rule

$$N_k = \left(\underline{N} \vee N_{k-1} - \epsilon \frac{\partial \Xi}{\partial N}(N_{k-1}, M_{k-1}) \right) \wedge \bar{N},$$

$$M_k = \left(\underline{M} \vee M_{k-1} - \epsilon \frac{\partial \Xi}{\partial M}(N_{k-1}, M_{k-1}) \right) \wedge \bar{M},$$

where $\epsilon > 0$.

- Select the optimal value $(N^*, M^*) = (N_{k^*}, M_{k^*})$, where k^* is the smallest k such that

$$\delta_k := \frac{|\Xi(N_k, M_k) - \Xi(N_{k-1}, M_{k-1})|}{\Xi(N_{k-1}, M_{k-1})} < \vartheta,$$

where $\vartheta > 0$.

- Instead, if $\delta_k > \vartheta \forall k$, make the selection $(N^*, M^*) = (N_K, M_K)$.

It is immediate to realize that this method for the selection of the cutting frequencies can be employed also when one is interested in implementing the Fourier spot covariance estimator, since, in that case, it is sufficient to replace $\Xi(N, M)$ with $\Phi(N, M)$ in the objective function, see Appendix D.

4.1.2. Efficiency. We evaluate the efficiency of the feasible method for selecting (N, M) in a simulation study.[†] We simulate a 2-dimensional noisy price process, in which asset 1 and asset 2 are supposed to represent, respectively, an equity stock and an equity market index. We generate 1000 trajectories with a horizon $T = 1$, corresponding to 6.5 hours.

As the data-generating process for the efficient price, we employ the model by Barndorff-Nielsen *et al.* (2011). For $j = 1, 2$, the model reads as

$$\begin{cases} p^j(t) = p^j(0) + \mu_j t + \int_0^t \sigma_j(s) dW_s^j \\ \sigma_j(t) = \exp\{\alpha_j + \delta_j \zeta_j(t)\} \\ \zeta_j(t) = \zeta_j(0) + \gamma_j \int_0^t \zeta_j(s) ds + Z_t^j \end{cases}, \quad (16)$$

where the W^1, W^2, Z^1 and the Z^2 are standard Brownian motions such that $\phi_W := d\langle W^1, W^2 \rangle_t / dt$, $\phi_Z := d\langle Z^1, Z^2 \rangle_t / dt$ and $\phi_{W,Z}^{j,j'} = d\langle W^j, Z^{j'} \rangle_t / dt, j, j' = 1, 2$.

For the simulation of the model, we set $(\mu_1, \alpha_1, \delta_1, \gamma_1) = (0.03, -1/3, 1/4, 1/40)$, $(\mu_2, \alpha_2, \delta_2, \gamma_2) = (0.01, -1/2, 1/6, 1/30)$ and $(\phi_W, \phi_Z, \phi_{W,Z}^{1,1}, \phi_{W,Z}^{1,2}, \phi_{W,Z}^{2,1}, \phi_{W,Z}^{2,2}) = (0.4, 0.5, -0.6, 0, 0, -0.3)$. Note that the choice of $\phi_{W,Z}^{1,1}$ and $\phi_{W,Z}^{2,2}$ reproduces the presence of leverage effects, while the selection of ϕ_W and ϕ_Z induces a positive correlation between the prices of the

Table 1. MISE values and average selection of N and M for spot beta estimates obtained via (7) in the feasible and unfeasible cases.

v_1, v_2	Feasible		Unfeasible	
	N, M	MISE	N, M	MISE
1.5, 1	843, 3	$6.118 \cdot 10^{-3}$	817, 3	$5.734 \cdot 10^{-3}$
2.5, 1	774, 3	$6.438 \cdot 10^{-3}$	740, 3	$6.069 \cdot 10^{-3}$

two assets and the volatilities of the two assets, respectively. We initialize the simulation with $p^1(0) = \log(40)$, $p^2(0) = \log(100)$, $\zeta_1(0) \sim \mathcal{N}(0, 8)$ and $\zeta_2(0) \sim \mathcal{N}(0, 10)$.

Furthermore, for the noise data-generating process, we assume that η^1 and η^2 are two i.i.d. Gaussian sequences, independent of each other and the efficient log-prices p^1 and p^2 , with mean equal to zero and standard deviation equal to $s_j^\eta = v_j s_j^p$, $j = 1, 2$, where v_j denotes the noise-to-signal and s_j^p is the sample standard deviation of the tick-by-tick efficient return. For the simulation, we set $v_2 = 1$, while for v_1 we consider two alternatives, that is, $v_1 = 1.5$ and $v_1 = 2.5$.

Finally, with regard to the sampling scheme, we let observation times follow homogeneous Poisson processes with intensities $\lambda_1 = 2/3$ and $\lambda_2 = 1$, corresponding to, respectively, 0.66 and 1 observations per second, on average. A study of the sensitivity to the pair (λ_1, λ_2) of the spot covariance estimator in (3), which appears in the numerator of (7), is given in section 4.3.1.

Using the feasible adaptive method for selecting N and M illustrated in section 4.1, we reconstruct the spot beta path from simulated log-prices on the equally-spaced grid $\tau_i = iT/m, i = 0, \dots, m-1$, where $m = 13$ corresponds to a mesh size of 30 minutes.[‡] As measures of path-wise global estimation efficiency, we employ the Mean Integrated Squared Error (MISE), that is,

$$MISE := \frac{T}{m} \sum_{i=1}^m E \left[\left(\tilde{\beta}_{n,N,M}^{1,2}(\tau_i) - \beta^{1,2}(\tau_i) \right)^2 \right].$$

Table 1 shows the MISE values resulting from the application of the feasible method of section 4.1 for selecting the cutting frequencies N and M and provides a comparison with the MISE values obtained in the case of the unfeasible optimal selection. We observe that when asset 1 is noisier, that is, v_1 is larger, the optimal unfeasible value of N is reduced, which is to be expected given the role of the cutting frequency N in filtering out the noise. Instead, the optimal unfeasible value of M equals 3 for both values of v_1 considered. Finally, we note that the feasible method provides very satisfactory MISE values, compared to the unfeasible optimal case, by selecting the same M ($M = 3$) while slightly overestimating N , for both values of v_1 .

[†] The implementation of the method requires selecting the constraint region R and the vector of hyperparameters (ϵ, ϑ, K) . In the following, we set $R = [[10\rho(n)^{-1/2}], [30\rho(n)^{-1/2}]] \times [[0.15\rho(n)^{-1/4}], [0.75\rho(n)^{-1/4}]]$ and $(\epsilon, \vartheta, K) = (5 \cdot 10^4, 10^{-5}, 10^4)$.

[‡] A MATLAB code for the implementation of the Fourier spot covariance estimator is available within the FMVol library developed by Sanfelici and Toscano (2024).

4.2. Robustness

In this section, we evaluate the finite-sample robustness of the spot beta estimator in (7) to different features of the efficient microstructure noise and the volatility that augment the data-generating process considered in section 4.1.2. Hereafter, we refer to the latter as the baseline scenario.

4.2.1. Microstructure noise. We consider the following three scenarios for microstructure noise.

SCENARIO I In the first scenario, we modify the baseline data-generating process of section 4.1.2 by adopting an inhomogeneous Poisson process for sampling observation times. Specifically, we set the sampling rates for the two assets equal to, respectively, $\lambda_1(t) = 15600(1 + \frac{1}{2}\cos(\frac{2\pi}{T}t))$ and $\lambda_2(t) = 23400(1 + \frac{1}{2}\cos(\frac{2\pi}{T}t))$. This sampling scheme attempts to reproduce the U-shaped pattern often observed in intra-day volatilities, which is related to the higher number of orders usually placed near the market open and close (see Jacod *et al.* 2017).

SCENARIO II In the second scenario, while maintaining the same sampling scheme and efficient price process as in the first scenario, we replace the i.i.d. Gaussian noise specification with the autocorrelated and price-dependent structure used in Jacod *et al.* (2017). For $j = 1, 2$ and $i = 0, 1, \dots, n_j$, using the simplified notation $\eta_i^j := \eta_{t_i}^j$, the latter reads as

$$\eta_i^j = \psi_i^j \chi_i^j,$$

where χ_i satisfies

$$\chi_i^j = Z_i^j + \sum_{l=1}^{L_j} \frac{d_j(1+d_j) \cdots (l-1+d_j)}{l!} Z_{i-l}^j, \\ d_j \in (-0.5, 0.5), \quad Z_i^j \sim_{i.i.d.} \mathcal{N}(0, z_j), \quad (17)$$

and ψ_i is sampled from

$$d\psi^j(t) = u_j(h_j(t) - \psi^j(t))dt + v_j dW_t^j, \\ h_j(t) = 1 + w_j \cos\left(\frac{2\pi}{T}t\right), \quad (18)$$

with W^1 and W^2 being the Brownian motions driving the dynamics of the efficient prices p^1 and p^2 .

The model in (17) is a moving-average series that approximates a fractionally differenced process. Specifically, in the limit $L_j \rightarrow \infty$, one obtains a fractionally differenced process whose autocorrelation function has a polynomial decay with rate $2d_j - 1$. This modeling choice attempts to replicate the slow-decay autocorrelations in the noise process empirically observed by Jacod *et al.* (2017). The model in (18) is an Ornstein-Uhlenbeck process that captures the dependence between the price and the noise of the j th asset by assuming the same Brownian driver W^j . Further, the deterministic drift h contributes to generating a U-shaped intraday volatility pattern.

For the simulation of the noise, we employ two alternative parameter selections, namely:

$$(z_1, d_1, L_1, u_1, w_1, v_1) = ((s_A^p \cdot 1.330)^2, 0.3, 100, 10, 0.3, 0.5), \\ (z_2, d_2, L_2, u_2, w_2, v_2) = ((s_V^p \cdot 0.887)^2, 0.3, 100, 10, 0.3, 0.5),$$

and

$$(z_1, d_1, L_1, u_1, w_1, v_1) = ((s_A^p \cdot 2.217)^2, 0.3, 100, 10, 0.3, 0.5), \\ (z_2, d_2, L_2, u_2, w_2, v_2) = ((s_V^p \cdot 0.887)^2, 0.3, 100, 10, 0.3, 0.5).$$

On average, the resulting noise-to-signal ratios (v_1, v_2) are roughly equal to $(1.5, 1)$ and $(2.5, 1)$, respectively. For all simulated trajectories, the initial conditions are chosen as $\psi_0^1 = \psi_0^2 = 1.3$.

SCENARIO III The third scenario augments the second with the introduction of price rounding. Specifically, for $j = 1, 2$ and $i = 0, 1, \dots, n_j$, the observed price process is given by

$$\tilde{p}_{t_i}^j = \log\left(\left[\exp\{p_{t_i}^j + \eta_{t_i}^j\}/\omega_j\right] \cdot \omega_j\right),$$

where $[x]$ denotes the integer part of $x \in \mathbb{R}$ and ω_j represents the tick size of the j th asset. We set $\omega_1 = \omega_2 = 0.01$.

Table 2 compares feasible and unfeasible MISE values in the three scenarios considered. We note that the optimal unfeasible values of N are reduced, compared to the baseline case in table 1, for both values of v_1 considered. Specifically, we observe a progressive reduction in N when moving from one scenario to the next. However, it is worth noting that such reduction is more pronounced when moving from the baseline scenario to Scenario I and from Scenarios I to II (especially when v_1 equals 2.5). This suggests that the introduction of inhomogeneous sampling (Scenario I) and auto-correlated and price-dependent noise (Scenario II) has a greater impact on the reduction of N , compared to rounding (Scenario III). As for M , its optimal value equals 3 in all scenarios, independently of the values of v_1 considered, and is unchanged with respect to the baseline scenario. Finally, we note that the feasible method for selecting the cutting frequencies is quite robust to alternative scenarios, as it still provides very satisfactory accuracy, compared to the unfeasible case.

4.2.2. Rough volatility. As detailed in Gatheral *et al.* (2018), rough models can be more accurate in reproducing the empirical behavior of volatility, compared to Brownian models. Accordingly, in this section, we employ a bivariate version of the Rough Heston model by El Euch and Rosenbaum (2019), for the efficient log-price, which reads as

$$\begin{cases} p^j(t) = p^j(0) + \int_0^t (\mu_j - \frac{1}{2}\sigma_j^2(s))ds + \int_0^t \sigma_j(s)dW_s^j \\ \sigma_j^2(t) = \sigma_j^2(0) + \theta_j \int_0^t (t-s)^{H_j-1/2} (\alpha_j - \sigma_j^2(s))ds \\ \quad + \zeta_j \int_0^t (t-s)^{H_j-1/2} \sigma_j(s)dZ_s^j \end{cases}, \quad (19)$$

where $H_j \in (0, 1/2)$ denotes the Hurst parameter and ϕ_W, ϕ_Z and $\phi_{W,Z}^{j,j'}, j' = 1, 2$, are defined as for the model in (16). For the simulation, we choose $H_1 = H_2 = 0.2$, $(\mu_1, \theta_1, \alpha_1, \zeta_1) = (0.05, 0.5, 0.4, 0.2)$ and $(\mu_2, \theta_2, \alpha_2, \zeta_2) = (0.02, 0.7, 0.3, 0.1)$.

Table 2. MISE values and average selection of N and M for spot beta estimates obtained via (7) in the feasible and unfeasible cases, for the different scenarios considered.

Scenario	v_1, v_2	Feasible		Unfeasible	
		N, M	MISE	N, M	MISE
I	1.5, 1	743, 3	$7.282 \cdot 10^{-3}$	763, 3	$7.008 \cdot 10^{-3}$
	2.5, 1	714, 3	$7.918 \cdot 10^{-3}$	636, 3	$7.619 \cdot 10^{-3}$
II	1.5, 1	720, 3	$7.433 \cdot 10^{-3}$	734, 3	$7.102 \cdot 10^{-3}$
	2.5, 1	650, 3	$8.187 \cdot 10^{-3}$	628, 3	$7.630 \cdot 10^{-3}$
III	1.5, 1	702, 3	$7.564 \cdot 10^{-3}$	733, 3	$7.171 \cdot 10^{-3}$
	2.5, 1	647, 3	$8.229 \cdot 10^{-3}$	626, 3	$7.652 \cdot 10^{-3}$

For the correlation parameters $(\phi_W, \phi_Z, \phi_{W,Z}^{1,1}, \phi_{W,Z}^{1,2}, \phi_{W,Z}^{2,1}, \phi_{W,Z}^{2,2})$, we select the same values as in the case of the simulation of the model in (16). For the initial price and volatility values, we make the selection $p^1(0) = \log(40)$, $p^2(0) = \log(100)$, $\sigma_1^2(0) = \alpha_1$ and $\sigma_2^2(0) = \alpha_2$.

Table 3 shows feasible and unfeasible MISE values of the spot beta estimator in (7) when the efficient price follows the model in (19), in the different scenarios considered for the noise.[†] As in the case with Brownian volatility in (16), the optimal value of the cutting frequency N is progressively reduced only when moving from one scenario to the next, while the cutting frequency M remains stable and equal to 3 in all scenarios considered. Furthermore, we note that the feasible method for selecting the cutting frequencies is also robust in the presence of rough volatility, since MISE values in table 3 values are again quite satisfactory, compared to the unfeasible case.

4.3. Performance comparison

In this section, we compare the finite-sample performance of the spot beta estimator in (7) with those of the LMM and realized spot beta estimators. As a preliminary study, we analyze the impact of the relative liquidity of the equity asset with respect to the market on the Fourier spot covariance estimator in (3), the LMM spot covariance estimator by Bibinger *et al.* (2019) and the spot realized covariance by Aït-Sahalia and Jacod (2014), on which the estimator in (7), the LMM spot beta estimator and realized spot beta estimator (see equation (5) in Andersen *et al.* 2021) are based, respectively. The reader may refer to section 3 of Bibinger *et al.* (2019) and Chapter 9 of Aït-Sahalia and Jacod (2014) for the definitions of the estimators and the illustration of their properties. We note that the LMM estimator is robust to the presence of noisy asynchronous observations, while the spot realized covariance is not. However, it is employed in our comparison study as a benchmark to quantify the performance gains of robust estimators like Fourier and LMM at high frequencies.

4.3.1. Measuring the impact of liquidity on spot covariance estimation. For the simulation of different liquidity profiles, we adopt the baseline scenario described in

section 4.1.2, and let the Poisson intensity for the equity asset λ_1 decrease while the one for the market index λ_2 is kept fixed and equal to 1. This way, we generate different trading volumes for the equity asset, such that the average number of daily observations relative to the market index is progressively reduced. Specifically, we consider the pairs $(\lambda_1, \lambda_2) \in \{(1, 1), (\frac{2}{3}, 1), (\frac{1}{3}, 1), (\frac{1}{5}, 1), (\frac{1}{10}, 1)\}$; on average, these values of (λ_1, λ_2) generate 23400 market index observations and, respectively, 23400, 15600, 7800, 4680 and 2340 equity asset observations, per daily trajectory. In the case when $(\lambda_1, \lambda_2) = (1, 1)$, we further assume that observations are synchronous on the 1-second equally-spaced grid. This way, the case $(\lambda_1, \lambda_2) = (1, 1)$ provides a benchmark to appreciate the effect of asynchronous sampling on finite-sample performance in the cases when $\lambda_1 < \lambda_2$.

Spot covariance paths are reconstructed on the 30-minute equally-spaced grid. For all estimators, we conduct an unfeasible optimization of the tuning parameters involved in the implementation thereof, to demonstrate the best possible performance. Tables 4 and 5 show the results of the performance comparison under different liquidity conditions. For a more comprehensive performance evaluation, in addition to MISE values, the tables include also Mean Integrated Error (MIE) values.[‡]

The comparison of MIE values suggests that the Fourier estimator outperforms the LMM estimator in terms of finite-sample integrated bias for all pairs (λ_1, λ_2) considered. However, we note that the MIE gap between the Fourier and LMM estimators is reduced when $v_1 = 2.5$, compared to the case when $v_1 = 1.5$, especially when λ_1 is smaller. Moreover, we observe that the naive tick-by-tick realized covariance carries a very large finite-sample integrated bias. This is particularly evident in the presence of asynchronous sampling, i.e. when $\lambda_1 < \lambda_2$. Indeed, under this condition, MIE values of the tick-by-tick realized covariance are larger by two orders of magnitude, compared to those of the Fourier and LMM estimators. This can be explained by the fact that the naive tick-by-tick realized estimator is affected by the well-known Epps effect (see Epps 1979), according to which covariance estimates are pushed towards zero as the sampling frequencies are increased in the presence of asynchronicity and microstructure effects. Indeed, when the realized covariances are computed using

[†] The baseline scenario in the rough volatility case coincides with the one described in section 4.1.2, the only difference being that (19) replaces (16) as the model for the efficient price.

[‡] The spot covariance MIE and MISE are given by, respectively, $\frac{T}{m} \sum_{i=1}^m E[\tilde{\Sigma}_{n,N,M}^{1,2}(\tau_i) - \Sigma^{1,2}(\tau_i)]$ and $\frac{T}{m} \sum_{i=1}^m E[(\tilde{\Sigma}_{n,N,M}^{1,2}(\tau_i) - \Sigma^{1,2}(\tau_i))^2]$, where $\tau_i = iT/m, i = 0, \dots, m-1$. For 30-minute estimates, m is equal to 13.

Table 3. MISE values and average selection of N and M for spot beta estimates obtained via (7) in the feasible and unfeasible cases when the efficient price follows the rough volatility model in (19), for the different scenarios considered.

Scenario	ν_1, ν_2	Feasible		Unfeasible	
		N, M	MISE	N, M	MISE
Baseline (rough)	1.5, 1	766, 3	$3.544 \cdot 10^{-3}$	818, 3	$3.379 \cdot 10^{-3}$
	2.5, 1	747, 3	$4.058 \cdot 10^{-3}$	740, 3	$3.531 \cdot 10^{-3}$
I	1.5, 1	742, 3	$3.878 \cdot 10^{-3}$	739, 3	$3.622 \cdot 10^{-3}$
	2.5, 1	647, 3	$4.321 \cdot 10^{-3}$	650, 3	$3.750 \cdot 10^{-3}$
II	1.5, 1	708, 3	$4.108 \cdot 10^{-3}$	715, 3	$3.942 \cdot 10^{-3}$
	2.5, 1	658, 3	$4.365 \cdot 10^{-3}$	618, 3	$4.096 \cdot 10^{-3}$
III	1.5, 1	706, 3	$4.109 \cdot 10^{-3}$	714, 3	$3.963 \cdot 10^{-3}$
	2.5, 1	657, 3	$4.421 \cdot 10^{-3}$	616, 3	$4.097 \cdot 10^{-3}$

Table 4. MIE and MISE values for the Fourier, LMM, and realized spot covariance estimators for different values of the pair (λ_1, λ_2) in the baseline scenario described in section 4.1.2, with $\nu_1 = 1.5$.

(λ_1, λ_2)	(1, 1)	$(\frac{2}{3}, 1)$	$(\frac{1}{3}, 1)$	$(\frac{1}{5}, 1)$	$(\frac{1}{10}, 1)$
	MIE				
Fourier	$2.567 \cdot 10^{-4}$	$-1.649 \cdot 10^{-3}$	$-3.918 \cdot 10^{-3}$	$-4.312 \cdot 10^{-3}$	$-7.526 \cdot 10^{-3}$
LMM	$7.694 \cdot 10^{-4}$	$-4.092 \cdot 10^{-3}$	$-5.880 \cdot 10^{-3}$	$-8.786 \cdot 10^{-3}$	$-1.466 \cdot 10^{-2}$
Real. Cov. (tick-by-tick)	$3.853 \cdot 10^{-3}$	$-1.560 \cdot 10^{-1}$	$-1.567 \cdot 10^{-1}$	$-1.603 \cdot 10^{-1}$	$-1.621 \cdot 10^{-1}$
Real. Cov. (3 min.)	$-3.041 \cdot 10^{-3}$	$-8.740 \cdot 10^{-3}$	$-1.202 \cdot 10^{-2}$	$-1.634 \cdot 10^{-2}$	$-2.893 \cdot 10^{-2}$
	MISE				
Fourier	$5.594 \cdot 10^{-4}$	$8.238 \cdot 10^{-4}$	$1.032 \cdot 10^{-3}$	$1.405 \cdot 10^{-3}$	$1.640 \cdot 10^{-3}$
LMM	$6.943 \cdot 10^{-4}$	$9.285 \cdot 10^{-4}$	$1.211 \cdot 10^{-3}$	$1.651 \cdot 10^{-3}$	$1.912 \cdot 10^{-3}$
Real. Cov. (tick-by-tick)	$3.427 \cdot 10^{-2}$	$4.208 \cdot 10^{-2}$	$4.489 \cdot 10^{-2}$	$4.544 \cdot 10^{-2}$	$4.765 \cdot 10^{-2}$
Real. Cov. (3 min.)	$1.245 \cdot 10^{-2}$	$1.652 \cdot 10^{-2}$	$1.659 \cdot 10^{-2}$	$1.664 \cdot 10^{-2}$	$1.679 \cdot 10^{-2}$

sparsely sampled 3-minute synchronous prices, the associated MIE values improve, as sparse sampling on a synchronous grid mitigates the effects of asynchronicity and noise.

As for MISE values, similar considerations as for MIE values apply. The Fourier estimator still shows a competitive edge with respect to the LMM estimator for all pairs (λ_1, λ_2) considered, even though the performance gap is reduced as λ_1 decreases. Furthermore, the naive spot realized covariance remains inefficient relative to the Fourier and LMM estimators based on MISE values, even when 3-minute observations are used.

4.3.2. Spot beta estimation. The efficiency of the Fourier spot covariance estimator relative to alternative estimation methods is reflected in a performance gain in the estimation of spot beta paths. Tables 6 and 7 provide MISE values for

the LMM and realized spot beta estimators under the different scenarios when the efficient price follows, respectively, the Brownian model in (16) and the rough volatility model in (19). Again, the LMM and realized estimator have been implemented via an unfeasible MISE-optimal selection of the tuning parameters. Further, note that the realized spot beta estimator has not been computed with tick-by-tick prices but with 3-minute prices, based on the findings about the finite-sample performance of the spot realized covariance estimator from the previous section.

By comparing MISE values in table 6 with those in tables 1 and 2, we observe that the spot beta estimator in (7) outperforms the LMM and realized spot beta estimators under the Brownian model in (16), and this happens not only in the case of the unfeasible selection of the cutting frequencies but also in the feasible case. However, it is noticeable that while the realized estimator appears to be inefficient, with MISE values

Table 5. MIE and MISE values for the Fourier, LMM, and realized spot covariance estimators for different values of the pair (λ_1, λ_2) in the baseline scenario described in section 4.1.2, with $\nu_1 = 2.5$.

(λ_1, λ_2)	(1, 1)	$(\frac{2}{3}, 1)$	$(\frac{1}{3}, 1)$	$(\frac{1}{5}, 1)$	$(\frac{1}{10}, 1)$
	MIE				
Fourier	$3.989 \cdot 10^{-4}$	$-2.491 \cdot 10^{-3}$	$-6.030 \cdot 10^{-3}$	$-9.115 \cdot 10^{-3}$	$-2.342 \cdot 10^{-2}$
LMM	$8.623 \cdot 10^{-4}$	$-4.502 \cdot 10^{-3}$	$-6.750 \cdot 10^{-3}$	$-9.350 \cdot 10^{-3}$	$-2.390 \cdot 10^{-2}$
Real. Cov. (tick-by-tick)	$4.738 \cdot 10^{-3}$	$-1.580 \cdot 10^{-1}$	$-1.605 \cdot 10^{-1}$	$-1.623 \cdot 10^{-1}$	$-1.627 \cdot 10^{-1}$
Real. Cov. (3 min.)	$-3.085 \cdot 10^{-3}$	$-1.262 \cdot 10^{-2}$	$-1.991 \cdot 10^{-2}$	$-2.969 \cdot 10^{-2}$	$-5.528 \cdot 10^{-2}$
	MISE				
Fourier	$5.814 \cdot 10^{-4}$	$8.506 \cdot 10^{-4}$	$1.109 \cdot 10^{-3}$	$1.586 \cdot 10^{-3}$	$2.411 \cdot 10^{-3}$
LMM	$7.495 \cdot 10^{-4}$	$9.904 \cdot 10^{-4}$	$1.404 \cdot 10^{-3}$	$1.797 \cdot 10^{-3}$	$2.669 \cdot 10^{-3}$
Real. Cov. (tick-by-tick)	$4.512 \cdot 10^{-2}$	$4.690 \cdot 10^{-2}$	$4.816 \cdot 10^{-2}$	$5.347 \cdot 10^{-2}$	$5.451 \cdot 10^{-1}$
Real. Cov. (3 min.)	$1.387 \cdot 10^{-2}$	$1.441 \cdot 10^{-2}$	$1.720 \cdot 10^{-2}$	$1.732 \cdot 10^{-2}$	$1.762 \cdot 10^{-2}$

Table 6. MISE values for the spot LMM and Realized beta estimators in the different scenarios when the efficient price follows the Brownian stochastic volatility model in (16).

Scenario	ν_1, ν_2	LMM	Realized Beta (3 min.)
Baseline	1.5, 1	$7.599 \cdot 10^{-3}$	$1.008 \cdot 10^{-1}$
	2.5, 1	$8.207 \cdot 10^{-3}$	$1.202 \cdot 10^{-1}$
I	1.5, 1	$7.716 \cdot 10^{-3}$	$1.114 \cdot 10^{-1}$
	2.5, 1	$8.547 \cdot 10^{-3}$	$1.233 \cdot 10^{-1}$
II	1.5, 1	$7.789 \cdot 10^{-3}$	$1.244 \cdot 10^{-1}$
	2.5, 1	$9.058 \cdot 10^{-3}$	$1.442 \cdot 10^{-1}$
III	1.5, 1	$7.900 \cdot 10^{-3}$	$1.245 \cdot 10^{-1}$
	2.5, 1	$9.064 \cdot 10^{-3}$	$1.442 \cdot 10^{-1}$

Table 7. MISE values for the spot LMM and Realized beta estimators in the different scenarios when the efficient price follows the rough volatility model in (19).

Scenario	ν_1, ν_2	LMM	Realized Beta (3 min.)
Baseline (rough)	1.5, 1	$4.021 \cdot 10^{-3}$	$5.071 \cdot 10^{-2}$
	2.5, 1	$4.368 \cdot 10^{-3}$	$5.524 \cdot 10^{-2}$
I	1.5, 1	$4.069 \cdot 10^{-3}$	$5.233 \cdot 10^{-2}$
	2.5, 1	$4.499 \cdot 10^{-3}$	$5.618 \cdot 10^{-2}$
II	1.5, 1	$4.185 \cdot 10^{-3}$	$5.345 \cdot 10^{-2}$
	2.5, 1	$4.662 \cdot 10^{-3}$	$5.622 \cdot 10^{-2}$
III	1.5, 1	$4.187 \cdot 10^{-3}$	$5.349 \cdot 10^{-2}$
	2.5, 1	$4.665 \cdot 10^{-3}$	$5.625 \cdot 10^{-2}$

that are two orders of magnitudes larger, the LMM estimator provides much more competitive MISE values of the same order of magnitude. Similar considerations apply when comparing tables 7 with 3, that is, in the case when the rough volatility model in (19) is employed. Nonetheless, in this case the performance gap between the spot beta estimator in (7) and competing estimators appears to be reduced. In particular, the LMM estimator yields MISE values that are closer to those produced by the estimator in (7), while the realized estimator improves its performance gap by one order of magnitude.

5. Empirical study

In this section, we present an empirical study that shows the existence of a positive dependence between the intraday dynamics of the cross-sectional dispersion of spot betas and the cross-sectional mean and standard deviation of the variance of the microstructure noise process.

The extensive empirical evidence produced by Andersen *et al.* (2021) shows that - contrary to what is commonly assumed in the estimation of asset pricing models - stock betas are not constant within a trading day. Furthermore, Andersen *et al.* (2021) study the cross-sectional dispersion of spot betas at time $t \in (0, T)$, that is,

$$D(t) := \frac{1}{d} \sum_{j=1}^d (\hat{\beta}^j(t) - 1)^2, \quad (20)$$

where $\hat{\beta}^j(t)$ indicates the spot beta estimate for the j th stock, $j = 1, \dots, d$, and obtain empirical evidence showing that the

$D(t)$ changes during the trading day, with paths that vary over time. Our empirical study aims to provide an economic rationale that may link intraday variations in the spot beta dispersion to intraday microstructure noise patterns and produce statistically significant results that support it.

As pointed out by Aït-Sahalia and Yu (2009), a reduced microstructure noise variance is typically associated with enhanced liquidity, reflected in narrower bid-ask spreads and/or ampler market depth. Improved liquidity, in turn, translates to lower transaction costs, subsequently diminishing the expected return on investment, and thus possibly reducing the cost of equity via a decrease in the beta. Moreover, less intense microstructure noise can indicate heightened efficiency and decreased informational asymmetry, leading to faster dissemination of information and quicker price adjustments in response to new information. This, in turn, might contribute to a lower perceived risk, accompanied by a decrease in its idiosyncratic component, namely the asset beta. For the specific case of beta cross-sectional dispersion, we argue that the latter may be positively dependent on the cross-sectional mean and standard deviation of the microstructure noise variance. Near market opening and closing, when microstructure noise tends to be larger and more volatile across assets (see, e.g. Andersen *et al.* 2022, and the references therein), more pronounced differences in liquidity and informational asymmetry across assets may generate a larger dispersion in betas. On the contrary, during the rest of the trading day, beta values may be more likely to be close to 1.

We test the existence of a positive dependence at the intraday level between the cross-sectional dispersion of spot betas and cross-sectional mean and standard deviation of the microstructure noise variance using tick-by-tick prices of 100 US stocks,[†] sampled over the period January 1, 2023–December 31, 2023, corresponding to 248 full trading days.[‡]

We begin with estimating the intraday path of spot betas using the estimator given in (7). As an empirical proxy for the market index, we employ the S&P500 index (SPX). We implement the spot beta estimator with tick-by-tick prices.[§] For each stock, we obtain beta estimates every 30 minutes during the time interval from 9.30 a.m. to 4 p.m., that is, for the k th day in the sample, we set $T = 1$ and adopt the estimation grid $\tau_i^k = iT/13, i = 0, 1, \dots, 13, k = 1, \dots, 248$. Note that we replace τ_0^k with $\tau_0^k + 5$ minutes and τ_{13}^k with $\tau_{13}^k - 5$ minutes to avoid boundary estimates. As for the selection of the cutting frequencies, we employ the feasible method illustrated in section 4.1.

The empirical characteristics of the intraday patterns of the reconstructed spot betas suggest that peaks, in absolute value, are usually observed near the market opening and closing. As an example, figure 1 shows the estimated daily spot beta paths on January 1, 2023, for two of the stocks included in

[†] The list of the stocks is provided in Appendix E.

[‡] Days with early closure were discarded.

[§] We test for jumps in tick-by-tick log-price increments of the stocks and the index using the test by Lee and Mykland (2012). We replace each jump with a zero log return. Overnight returns, which typically contain jumps resulting from the release of company news during market closures, are excluded from the study.

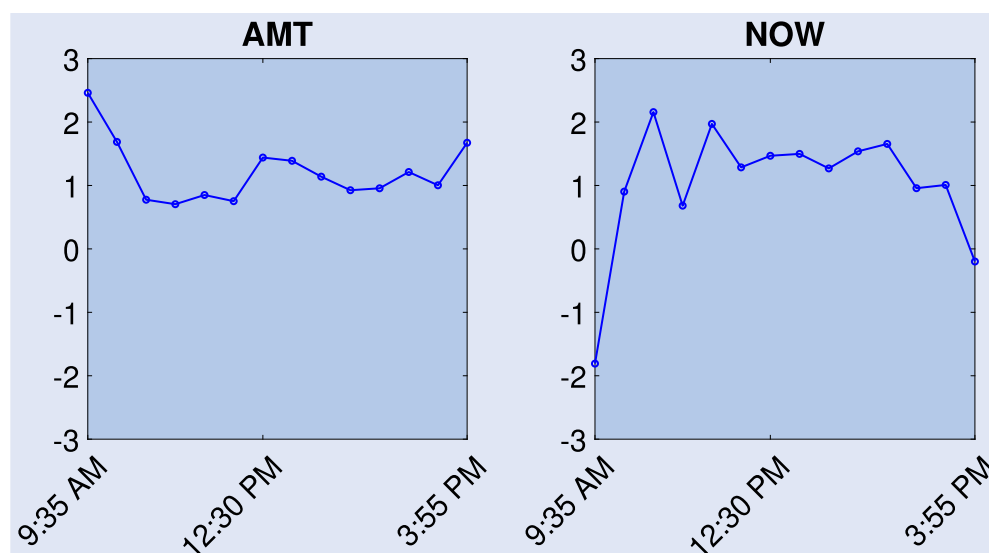


Figure 1. Reconstructed intraday spot beta paths for AMT and NOW on January 1, 2023.

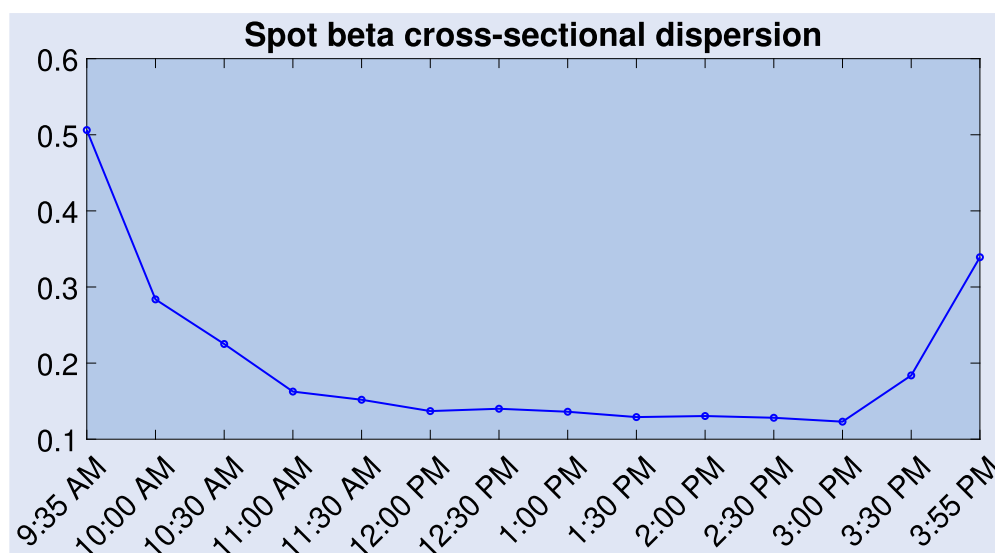


Figure 2. Intraday cross-sectional dispersion (20) for the 100 stocks employed in our study on January 1, 2023. Spot beta estimates were computed with the estimator in (7), implemented with tick-by-tick prices.

the study, namely AMT and NOW. Additionally, figure 2 provides the intraday path on January 1, 2023, of the dispersion measure in (20) implemented with the spot beta estimates obtained via (7). Similarly to what is observed in Andersen *et al.* (2021), the plot shows a declining pattern until 3.00 p.m.[†] The largest daily values are however observed near the boundaries.

Once the time series of the intraday cross-sectional dispersion of spot betas have been reconstructed for all the days in the sample, we test their dependence on the cross-sectional average and standard deviation of the microstructure noise variance. Specifically, we estimate the following two

alternative log-linear models, that is,

$$\log D(\tau_i^k) = q_A A(\tau_i^k) + p_A \quad (21)$$

and

$$\log D(\tau_i^k) = q_V V(\tau_i^k) + p_V, \quad (22)$$

where, for $i = 1, \dots, 13$, and $k = 1, \dots, 248$, $A(\tau_i^k)$ and $V(\tau_i^k)$ denote the cross-sectional mean and standard deviation of the noise variance estimates, computed with (15) by using tick-by-tick price observations sampled on $[\tau_{i-1}, \tau_i]$.

We avoid jointly including A_d and V_d in the linear regression due to the presence of collinearity, the sample correlation between these two variables being equal to 0.881. For the estimation of (21) and (22), to reduce the impact of outliers, we employ Iteratively Reweighted Least Squares, with the ‘bisquare’ weight function (see Holland and Welsch 1977). Based on the results of the Ljung-Box test, conducted at

[†] It is worth noting that Andersen *et al.* (2021) produce dispersion estimates employing 3-minute synchronous price observations sampled over local windows and obtain intraday dispersion estimates only in the period between 10.30 a.m. and 3.00 p.m.

Table 8. Results of the estimation of the models in (21) and (22).

	Estimate	std error	p-value	R^2	AIC	BIC
$\hat{q}_A$	$5.383 \cdot 10^6$	$1.409 \cdot 10^6$	$1.326 \cdot 10^{-4}$			
$\hat{p}_A$	-1.222	$5.352 \cdot 10^{-2}$	0	0.171	$9.271 \cdot 10^3$	$9.283 \cdot 10^3$
$\hat{q}_V$	$3.550 \cdot 10^6$	$3.337 \cdot 10^5$	0			
$\hat{p}_V$	-1.221	$3.008 \cdot 10^{-2}$	0	0.331	$8.884 \cdot 10^3$	$8.896 \cdot 10^3$

the 5% significance level, residuals show evidence of autocorrelations. Hence, we compute Newey-West standard errors (see Newey and West 1987).

The results of the estimation are summarized in table 8. Based on p-values, the estimated coefficient pairs $(\hat{q}_A, \hat{p}_A)$ and $(\hat{q}_V, \hat{p}_V)$ are both statistically significant. Moreover, the positive sign of $\hat{q}_A$ and $\hat{q}_V$ supports the hypothesis of the existence of a positive dependence between the cross-sectional dispersion of spot betas and, respectively, the cross-sectional mean and standard deviation of the noise variance. However, the comparison of AIC and BIC values suggests that the cross-sectional standard deviation of the noise variance might be preferable as a predictor of beta dispersion. The same conclusion is also supported by the comparison of R^2 values.

6. Conclusions

This paper contributes to the study of non-parametric Fourier-based estimation of multivariate spot volatilities with realistic noisy and asynchronous data, both theoretically and empirically. Firstly, we prove the rate-efficient point-wise asymptotic normality for the estimation error of the Fourier spot covariance estimator in the presence of asynchronous observations and microstructure noise effects. Then, we exploit the latter result to obtain a consistent estimator of the spot beta. Moreover, we obtain numerical evidence suggesting that such an estimator outperforms the LMM and realized spot beta estimators in the presence of several realistic features, such as rough volatility, inhomogeneous asynchronous sampling, autocorrelated and price-dependent noise and price rounding. Finally, using a yearly sample of tick-by-tick prices of 100 US stocks, we empirically show the existence of an economically motivated link between the intraday patterns of the cross-sectional dispersion of spot betas and the cross-sectional average and standard deviation of the microstructure noise variance, used as an inverse proxy for the level of liquidity and informational efficiency.

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Appendices

A. Proofs

A.1. Proof of Theorem 3.1

REMARK A.1 As every continuous process is locally bounded, thus all processes in the proof are. Moreover, standard localization procedures (see, e.g., Aït-Sahalia and Jacod 2014) allow assuming that any locally bounded process is actually bounded and almost surely positive processes can be considered as bounded away from zero.

REMARK A.2 Given the discrete-time observations defined in Section 3, we use the notation in continuous time by letting $\varphi_n^1(t) := \sup\{t_i^1 : t_i^1 \leq t\}$ for the sake of simplicity; the same holds for $\varphi_n^2(t)$. Without loss of generality, we consider the interval $[0, 2\pi]$.

Proof According to Lemma 2.2 Malliavin and Mancino (2009), we can assume that the drift of both the price and the volatility processes are zero. For any $t \in (0, 2\pi)$ we rewrite the estimator error $\tilde{\Sigma}_{n,N,M}^{1,2}(t) - \Sigma^{1,2}(t)$ as:

$$A(t) + F_1(t) + F_2(t) + E_1(t) + E_2(t) + E_3(t), \quad (A1)$$

where

$$A(t) := \frac{1}{2\pi} \int_0^{2\pi} F_M(t - \varphi_n^2(v)) D_N(\varphi_n^1(v) - \varphi_n^2(v)) \Sigma^{1,2}(v) dv - \Sigma^{1,2}(t), \quad (A2)$$

$$F_1(t) := \frac{1}{2\pi} \int_0^{2\pi} F_M(t - \varphi_n^2(v)) \int_0^v D_N(\varphi_n^1(u) - \varphi_n^2(v)) dp^1(u) \times dp^2(v), \quad (A3)$$

$$F_2(t) := \frac{1}{2\pi} \int_0^{2\pi} \int_0^v F_M(t - \varphi_n^2(u)) D_N(\varphi_n^2(u) - \varphi_n^1(v)) dp^2(u) \times dp^1(v), \quad (A4)$$

$$E_1(t) := \frac{1}{2\pi} \sum_{i=0}^{n_1-1} \sum_{j=0}^{n_2-1} F_M(t - t_j^2) D_N(t_i^1 - t_j^2) \delta_i(p^1) \varepsilon_j^2, \quad (A5)$$

$$E_2(t) := \frac{1}{2\pi} \sum_{i=0}^{n_1-1} \sum_{j=0}^{n_2-1} F_M(t - t_j^2) D_N(t_i^1 - t_j^2) \delta_j(p^2) \varepsilon_i^1, \quad (A6)$$

$$E_3(t) := \frac{1}{2\pi} \sum_{i=0}^{n_1-1} \sum_{j=0}^{n_2-1} F_M(t - t_j^2) D_N(t_i^1 - t_j^2) \varepsilon_i^1 \varepsilon_j^2. \quad (A7)$$

■

A.1.1. No noise component. (I) In the first step we consider the term $A(t)$:

$$\rho(n)^{-1/4} M^{-1/2} \left(\frac{1}{2\pi} \int_0^{2\pi} F_M(t - \varphi_n^2(v)) D_N(\varphi_n^1(v) - \varphi_n^2(v)) \Sigma^{1,2}(v) dv - \Sigma^{1,2}(t) \right). \quad (A8)$$

The term (A8) is decomposed into two addends:

$$\rho(n)^{-1/4} M^{-1/2} \left(\frac{1}{2\pi} \int_0^{2\pi} F_M(t - \varphi_n^2(v)) D_N(\varphi_n^1(v) - \varphi_n^2(v)) \times \Sigma^{1,2}(v) dv - \frac{1}{2\pi} \int_0^{2\pi} F_M(t - v) \Sigma^{1,2}(v) dv \right) \quad (\text{A9})$$

$$+ \rho(n)^{-1/4} M^{-1/2} \left(\frac{1}{2\pi} \int_0^{2\pi} F_M(t - v) \Sigma^{1,2}(v) dv - \Sigma^{1,2}(t) \right). \quad (\text{A10})$$

The term (A9) is the discretization error and converges to zero in probability; indeed, we have:

$$E \left[\left| \frac{1}{2\pi} \int_0^{2\pi} F_M(t - \varphi_n^2(v)) D_N(\varphi_n^1(v) - \varphi_n^2(v)) \Sigma^{1,2}(v) dv - \frac{1}{2\pi} \int_0^{2\pi} F_M(t - v) \Sigma^{1,2}(v) dv \right| \right] \leq E \left[\left| \frac{1}{2\pi} \int_0^{2\pi} F_M(t - \varphi_n^2(v)) (D_N(\varphi_n^1(v) - \varphi_n^2(v)) - 1) \times \Sigma^{1,2}(v) dv \right| \right] \quad (\text{A11})$$

$$+ E \left[\left| \frac{1}{2\pi} \int_0^{2\pi} F_M(t - \varphi_n^2(v)) \Sigma^{1,2}(v) dv - \frac{1}{2\pi} \int_0^{2\pi} F_M(t - v) \Sigma^{1,2}(v) dv \right| \right]. \quad (\text{A12})$$

Consider (A11), we have

$$E \left[\left| \frac{1}{2\pi} \int_0^{2\pi} F_M(t - \varphi_n^2(v)) (D_N(\varphi_n^1(v) - \varphi_n^2(v)) - 1) \times \Sigma^{1,2}(v) dv \right| \right] \leq \text{ess sup } \|\Sigma^{1,2}\|_\infty \frac{1}{2\pi} \int_0^{2\pi} F_M(t - \varphi_n^2(v)) \times |D_N(\varphi_n^1(v) - \varphi_n^2(v)) - 1| dv = C \frac{1}{2\pi} \int_0^{2\pi} F_M(t - \varphi_n^2(v)) dv O(N^2 \rho(n)^2) = O(N^2 \rho(n)^2),$$

because $|D_N(\varphi_n^1(v) - \varphi_n^2(v)) - 1| = O(N^2 \rho(n)^2)$ in virtue of the assumption (S.II).

Consider (A12), we have

$$E \left[\left| \frac{1}{2\pi} \int_0^{2\pi} F_M(t - \varphi_n^2(v)) - F_M(t - v) \right| \Sigma^{1,2}(v) dv \right] \leq \text{ess sup } \|\Sigma^{1,2}\|_\infty \left| \int_0^{2\pi} F_M(t - \varphi_n^2(v)) dv - \int_0^{2\pi} F_M(t - v) dv \right| = O(M \rho(n_2)),$$

by Lemma A.3(i).

In the sequel $\gamma_{1,2}(t)$ will denote the volatility of the covariance process $\Sigma^{1,2}(t)$.

Consider now the term (A10), which we denote by $G(t)$. $G(t)$ splits into two terms:

$$G_1(t) := \frac{1}{2\pi} \int_{[0, 2\pi] \cap \{|t-v| \leq \frac{2\pi}{M+1}\}} F_M(t - v) (\Sigma^{1,2}(v) - \Sigma^{1,2}(t)) dv \quad (\text{A13})$$

and

$$G_2(t) := \frac{1}{2\pi} \int_{[0, 2\pi] \cap \{|t-v| > \frac{2\pi}{M+1}\}} F_M(t - v) (\Sigma^{1,2}(v) - \Sigma^{1,2}(t)) dv. \quad (\text{A14})$$

As for the term $G_2(t)$ in (A14), it is shown that

$$\rho(n)^{-1/4} M^{-1/2} E \left[\left| \frac{1}{2\pi} \int_{[0, 2\pi] \cap \{|t-v| > \frac{2\pi}{M+1}\}} F_M(t - v) \times (\Sigma^{1,2}(v) - \Sigma^{1,2}(t)) dv \right| \right] \leq \rho(n)^{-1/4} M^{-1/2} \frac{1}{2\pi} \int_{[0, 2\pi] \cap \{|t-v| > \frac{2\pi}{M+1}\}} F_M(t - v) \times E \left[|\Sigma^{1,2}(v) - \Sigma^{1,2}(t)| \right] dv \leq \rho(n)^{-1/4} M^{-1/2} 2 \text{ess sup } \|\Sigma^{1,2}\|_\infty \times \frac{1}{2\pi} \int_{[0, 2\pi] \cap \{|t-v| > \frac{2\pi}{M+1}\}} F_M(t - v) dv \leq C \rho(n)^{-1/4} M^{-3/2} = o_p(1),$$

by lemma A.3(ii).

Consider $G_1(t)$ in (A13), which is equal to:

$$\sum_{|k| \leq M} \left(1 - \frac{|k|}{M+1} \right) e^{ikt} \frac{1}{2\pi} \times \int_{[0, 2\pi] \cap \{|t-v| \leq \frac{2\pi}{M+1}\}} e^{-ikv} (\Sigma^{1,2}(v) - \Sigma^{1,2}(t)) dv. \quad (\text{A15})$$

First, we prove that the term corresponding to $k = 0$ in (A15) is asymptotically negligible. It splits into

$$\frac{1}{2\pi} \int_{t - \frac{2\pi}{M+1}}^t (\Sigma^{1,2}(v) - \Sigma^{1,2}(t)) dv + \frac{1}{2\pi} \int_t^{t + \frac{2\pi}{M+1}} (\Sigma^{1,2}(v) - \Sigma^{1,2}(t)) dv.$$

For the first addend, applying the integration by parts, it holds:

$$\frac{1}{2\pi} \int_{t - \frac{2\pi}{M+1}}^t (\Sigma^{1,2}(v) - \Sigma^{1,2}(t)) dv = \frac{1}{2\pi} [v (\Sigma^{1,2}(v) - \Sigma^{1,2}(t))]_{t - \frac{2\pi}{M+1}}^t - \frac{1}{2\pi} \int_{t - \frac{2\pi}{M+1}}^t v d\Sigma_v^{1,2} = \frac{1}{2\pi} \int_{t - \frac{2\pi}{M+1}}^t \left(t - \frac{2\pi}{M+1} - v \right) d\Sigma_v^{1,2}. \quad (\text{A16})$$

The first term of the asymptotic variance is obtained by taking the quadratic variation of (A16). We get:

$$\rho(n)^{-1/2} M^{-1} \frac{1}{(2\pi)^2} \int_{t - \frac{2\pi}{M+1}}^t \left(t - \frac{2\pi}{M+1} - v \right)^2 \gamma_{1,2}^2(v) dv = \rho(n)^{-1/2} M^{-1} \frac{1}{(2\pi)^2} \int_{t - \frac{2\pi}{M+1}}^t \left(t - \frac{2\pi}{M+1} - v \right)^2 dv \times \left(\gamma_{1,2}^2 \left(t - \frac{2\pi}{M+1} \right) + O_p \left(\frac{1}{M+1} \right) \right) = \rho(n)^{-1/2} M^{-1} \frac{1}{(2\pi)^2} \frac{1}{3} \left(\frac{2\pi}{M+1} \right)^3 \times \left(\gamma_{1,2}^2 \left(t - \frac{2\pi}{M+1} \right) + O_p \left(\frac{1}{M+1} \right) \right) = O_p \left(\rho(n)^{-1/2} M^{-4} \right) = o_p(1).$$

The second addend is analogous.

Thus, we consider (A15) for $k \neq 0$. Using the integration by parts, it is equal to:

$$\sum_{0 < |k| \leq M} \left(1 - \frac{|k|}{M+1}\right) e^{ikt} \frac{1}{2\pi} \left[\frac{i}{k} e^{-ikv} (\Sigma^{1,2}(v) - \Sigma^{1,2}(t)) \right]_{t - \frac{2\pi}{M+1}}^{t + \frac{2\pi}{M+1}} \quad (\text{A17})$$

$$+ \sum_{0 < |k| \leq M} \left(1 - \frac{|k|}{M+1}\right) e^{ikt} \frac{1}{2\pi} \frac{i}{k} \times \int_{[0, 2\pi] \cap \{|t-v| \leq \frac{2\pi}{M+1}\}} e^{-ikv} d\Sigma_v^{1,2}. \quad (\text{A18})$$

Consider (A17). It is equal to:

$$\begin{aligned} & \sum_{0 < |k| \leq M} \left(1 - \frac{|k|}{M+1}\right) e^{ikt} \frac{1}{2\pi} \frac{i}{k} \left(e^{-ik(t + \frac{2\pi}{M+1})} \right. \\ & \quad \times \left(\Sigma^{1,2}\left(t + \frac{2\pi}{M+1}\right) - \Sigma^{1,2}(t) \right) \\ & \quad \left. - e^{-ik(t - \frac{2\pi}{M+1})} \left(\Sigma^{1,2}\left(t - \frac{2\pi}{M+1}\right) - \Sigma^{1,2}(t) \right) \right) \\ & = \sum_{0 < |k| \leq M} \left(1 - \frac{|k|}{M+1}\right) \frac{1}{2\pi} \frac{i}{k} \\ & \quad \times \left(e^{-ik \frac{2\pi}{M+1}} \int_t^{t + \frac{2\pi}{M+1}} d\Sigma_v^{1,2} + e^{ik \frac{2\pi}{M+1}} \int_{t - \frac{2\pi}{M+1}}^t d\Sigma_v^{1,2} \right). \quad (\text{A19}) \end{aligned}$$

For the first integral (the second one is analogous), we study the convergence in probability of:

$$\begin{aligned} & \rho(n)^{-1/2} M^{-1} \frac{1}{(2\pi)^2} \\ & \quad \times \left\langle \sum_{0 < |k| \leq M} \left(1 - \frac{|k|}{M+1}\right) \frac{1}{k} e^{-ik \frac{2\pi}{M+1}} \int_t^{t + \frac{2\pi}{M+1}} d\Sigma^{1,2}, \right. \\ & \quad \left. \sum_{0 < |k| \leq M} \left(1 - \frac{|k|}{M+1}\right) \frac{1}{k} e^{ik \frac{2\pi}{M+1}} \int_t^{t + \frac{2\pi}{M+1}} d\Sigma^{1,2} \right\rangle. \end{aligned}$$

The dominant term is studied as follows

$$\begin{aligned} & \rho(n)^{-1/2} M^{-1} \frac{1}{(2\pi)^2} \sum_{0 < |k| \leq M} \left(1 - \frac{|k|}{M+1}\right)^2 \\ & \quad \times \frac{1}{k^2} \int_t^{t + \frac{2\pi}{M+1}} \gamma_{1,2}(s) ds \\ & = \frac{1}{2\pi} \frac{1}{M+1} \rho(n)^{-1/2} M^{-1} \sum_{0 < |k| \leq M} \left(1 - \frac{|k|}{M+1}\right)^2 \\ & \quad \times \frac{1}{k^2} \left(\gamma_{1,2}(t) + O_p\left(\frac{1}{M+1}\right) \right). \quad (\text{A20}) \end{aligned}$$

It is easily seen that

$$\begin{aligned} & \lim_M \sum_{0 < |k| \leq M} \left(1 - \frac{|k|}{M+1}\right)^2 \frac{1}{k^2} \\ & = \lim_M \sum_{0 < |k| \leq M} \left(1 - \frac{|k|}{M+1}\right) \frac{1}{k^2} \\ & = \lim_M \sum_{0 < |k| \leq M} \frac{1}{k^2} = \frac{\pi^2}{3}. \end{aligned}$$

Further, by the condition $M^2 \rho(n)^{1/2} \sim a^2 c$, and the continuity of $\gamma_{1,2}(t)$, we conclude that (A20) converges to

$$\frac{1}{a^2 c} \frac{\pi}{6} \gamma_{1,2}(t).$$

Finally, noting that the covariance between the two stochastic integrals in (A19) is zero, the contribution of (A17) is equal to

$$\frac{1}{a^2 c} \frac{\pi}{3} \gamma_{1,2}(t).$$

Similarly, we deal with the term (A18), which is equal to:

$$\begin{aligned} & \frac{1}{2\pi} \sum_{0 < |k| \leq M} \left(1 - \frac{|k|}{M+1}\right) e^{ikt} \\ & \quad \times \frac{i}{k} \left(\int_{t - \frac{2\pi}{M+1}}^t e^{-ikv} d\Sigma_v^{1,2} + \int_t^{t + \frac{2\pi}{M+1}} e^{-ikv} d\Sigma_v^{1,2} \right). \end{aligned}$$

The convergence in probability for the bracket of the first addend, that is,

$$\begin{aligned} & \rho(n)^{-1/2} M^{-1} \frac{1}{(2\pi)^2} \\ & \quad \left\langle \sum_{0 < |k| \leq M} \left(1 - \frac{|k|}{M+1}\right) e^{ikt} \frac{1}{k} \int_{t - \frac{2\pi}{M+1}}^t e^{-ikv} d\Sigma_v^{1,2}, \right. \\ & \quad \left. \times \sum_{0 < |k| \leq M} \left(1 - \frac{|k|}{M+1}\right) e^{-ikt} \frac{1}{k} \int_{t - \frac{2\pi}{M+1}}^t e^{ikv} d\Sigma_v^{1,2} \right\rangle, \end{aligned}$$

reads as follows:

$$\begin{aligned} & \rho(n)^{-1/2} M^{-1} \frac{1}{(2\pi)^2} \sum_{0 < |k| \leq M} \left(1 - \frac{|k|}{M+1}\right)^2 \frac{1}{k^2} \int_{t - \frac{2\pi}{M+1}}^t \gamma_{1,2}^2(v) dv \\ & = \rho(n)^{-1/2} M^{-1} \frac{1}{(2\pi)^2} \sum_{0 < |k| \leq M} \left(1 - \frac{|k|}{M+1}\right)^2 \frac{1}{k^2} \frac{2\pi}{M+1} \\ & \quad \times \left(\gamma_{1,2}^2\left(t - \frac{2\pi}{M+1}\right) + O_p\left(\frac{1}{M+1}\right) \right) \\ & \rightarrow \frac{1}{a^2 c} \frac{\pi}{6} \gamma_{1,2}^2(t). \end{aligned}$$

The second addend is analogous.

Finally these four terms contribute to the variance with

$$\frac{2\pi}{3} \frac{1}{a^2 c} \gamma_{1,2}^2(t).$$

(II) The study of the bracket

$$\left\langle \rho(n)^{-1/4} M^{-1/2} (F_1(t) + F_2(t)), \rho(n)^{-1/4} M^{-1/2} (F_1(t) + F_2(t)) \right\rangle \quad (\text{A21})$$

is composed by four terms.

Consider first

$$\begin{aligned} & \rho(n)^{-1/2} M^{-1} \langle F_1(t), F_1(t) \rangle \\ & = \rho(n)^{-1/2} M^{-1} \frac{1}{(2\pi)^2} \int_0^{2\pi} F_M^2(t - \varphi_n^2(v)) \\ & \quad \times \left(\int_0^v D_N(\varphi_n^1(s) - \varphi_n^2(v)) \sigma_1^*(s) dW_s \right)^2 \Sigma^{2,2}(v) dv \\ & = \rho(n)^{-1/2} M^{-1} (F_{11}(t) + F_{12}(t)), \end{aligned}$$

where

$$F_{11}(t) := \frac{1}{(2\pi)^2} \int_0^{2\pi} F_M^2(t - \varphi_n^2(v)) \int_0^v D_N^2(\varphi_n^1(s) - \varphi_n^2(v)) \times \Sigma^{1,1}(s) ds \Sigma^{2,2}(v) dv, \quad (A22)$$

$$F_{12}(t) := \frac{1}{(2\pi)^2} \int_0^{2\pi} F_M^2(t - \varphi_n^2(v)) \times \int_0^v \left(\int_0^s D_N(\varphi_n^1(v) - \varphi_n^2(u)) \sigma_1^*(u) dW_u \right) \times D_N(\varphi_n^1(s) - \varphi_n^2(v)) \sigma_1^*(s) dW_s \Sigma^{2,2}(v) dv. \quad (A23)$$

The term $\rho(n)^{-1/2} M^{-1} F_{12}(t)$ converges to zero in probability. Noting that $\rho(n)^{-1/2} M^{-1} = \frac{1}{c} N M^{-1}$, for $F_{11}(t)$ we get:

$$\begin{aligned} & \left| \frac{1}{c} \frac{N}{M} F_{11}(t) - \frac{1}{6c} \Sigma^{1,1}(t) \Sigma^{2,2}(t) \right| \\ &= \left| \frac{1}{c} \frac{N}{M} \frac{1}{(2\pi)^2} \int_0^{2\pi} F_M^2(t - \varphi_n^2(v)) \int_0^v D_N^2(\varphi_n^1(s) - \varphi_n^2(v)) \right. \\ & \quad \times \Sigma^{1,1}(s) ds \Sigma^{2,2}(v) dv - \frac{1}{c} \frac{1}{6} \Sigma^{1,1}(t) \Sigma^{2,2}(t) \left| \right. \\ &\leq \frac{1}{c} \frac{1}{(2\pi)^2} \int_0^{2\pi} \frac{1}{M} F_M^2(t - \varphi_n^2(v)) \\ & \quad \times \left| N \int_0^v D_N^2(\varphi_n^1(s) - \varphi_n^2(v)) \Sigma^{1,1}(s) ds \right. \\ & \quad \left. - \frac{\pi}{2} \Sigma^{1,1}(v) \right| \Sigma^{2,2}(v) dv \\ & \quad + \frac{1}{c} \left| \frac{\pi}{2} \frac{1}{(2\pi)^2} \int_0^{2\pi} \frac{1}{M} F_M^2(t - \varphi_n^2(v)) \Sigma^{1,1}(t) \Sigma^{2,2}(v) dv \right. \\ & \quad \left. - \frac{1}{6} \Sigma^{1,1}(t) \Sigma^{2,2}(t) \right|. \end{aligned} \quad (A24)$$

Using Lemmas A.6 and A.3(iv) (A24) goes to 0. Analogously, (A25) converges to 0 by Lemma A.6. The term

$$\rho(n)^{-1/2} M \langle F_2(t), F_2(t) \rangle$$

is analogous. Therefore the sum of the two terms converges to

$$\frac{1}{3c} \Sigma^{1,1}(t) \Sigma^{2,2}(t).$$

Considering the term

$$\rho(n)^{-1/2} M (\langle F_1(t), F_2(t) \rangle + \langle F_2(t), F_1(t) \rangle)$$

with an analogous procedure we obtain that it converges to

$$\frac{1}{3c} \left(\Sigma^{1,2}(t) \right)^2.$$

A.1.2. Noise component. In this section, we compute the contribution to the asymptotic variance from the pure noise term (A7). Let $\varepsilon_{t_i}^j = \eta_{t_{i+1}}^j - \eta_{t_i}^j$ and use the short notation ε_i^j for $\varepsilon_{t_i}^j$. We observe that, neglecting end-effects that will be studied later, the term

$$\begin{aligned} & \sum_{i=0}^{n_1-1} \sum_{j=0}^{n_2-1} F_M(t - t_j^2) D_N(t_i^1 - t_j^2) \varepsilon_i^1 \varepsilon_j^2 \\ &= \sum_{i=0}^{n_1-1} \sum_{j=0}^{n_2-1} F_M(t - t_j^2) D_N(t_i^1 - t_j^2) \\ & \quad \times (\eta_{t_{i+1}}^1 \eta_{t_{j+1}}^2 - \eta_{t_i}^1 \eta_{t_{j+1}}^2 - \eta_{t_{i+1}}^1 \eta_j^2 + \eta_i^1 \eta_j^2) \end{aligned}$$

can be expressed as follows:

$$\begin{aligned} & \sum_{i=0}^{n_1-1} \sum_{j=0}^{n_2-1} F_M(t - t_j^2) D_N(t_i^1 - t_j^2) \eta_{t_{i+1}}^1 \eta_{t_{j+1}}^2 \\ & - \sum_{i=0}^{n_1-1} \sum_{j=0}^{n_2-1} F_M(t - t_j^2) D_N(t_{i+1}^1 - t_j^2) \eta_{t_{i+1}}^1 \eta_{t_{j+1}}^2 \\ & - \sum_{i=0}^{n_1-1} \sum_{j=0}^{n_2-1} F_M(t - t_{j+1}^2) D_N(t_i^1 - t_{j+1}^2) \eta_{t_{i+1}}^1 \eta_{t_{j+1}}^2 \\ & + \sum_{i=0}^{n_1-1} \sum_{j=0}^{n_2-1} F_M(t - t_{j+1}^2) D_N(t_{i+1}^1 - t_{j+1}^2) \eta_{t_{i+1}}^1 \eta_{t_{j+1}}^2, \end{aligned} \quad (A26)$$

$$\begin{aligned} & - \sum_{j=0}^{n_2-1} F_M(t - t_j^2) D_N(t_0^1 - t_j^2) \eta_0^1 \eta_{t_{j+1}}^2 \\ & + \sum_{j=0}^{n_2-1} F_M(t - t_j^2) D_N(t_n^1 - t_j^2) \eta_{t_n}^1 \eta_{t_{j+1}}^2 \\ & - \sum_{i=0}^{n_1-1} F_M(t - t_0^2) D_N(t_j^1 - t_0^2) \eta_{t_{i+1}}^1 \eta_0^2 \\ & + \sum_{i=0}^{n_1-1} F_M(t - t_n^2) D_N(t_j^1 - t_n^2) \eta_{t_{i+1}}^1 \eta_{t_n}^2 \\ & + \sum_{j=0}^{n_2-1} F_M(t - t_j^2) D_N(t_0^1 - t_j^2) \eta_0^1 \eta_j^2 \\ & - \sum_{j=0}^{n_2-1} F_M(t - t_j^2) D_N(t_n^1 - t_j^2) \eta_{t_n}^1 \eta_j^2 \\ & + \sum_{i=0}^{n_1-1} F_M(t - t_0^2) D_N(t_i^1 - t_0^2) \eta_i^1 \eta_0^2 \\ & - \sum_{i=0}^{n_1-1} F_M(t - t_n^2) D_N(t_i^1 - t_n^2) \eta_i^1 \eta_{t_n}^2. \end{aligned} \quad (A27)$$

plus the remaining end points equal to the following:

$$\begin{aligned} & - \sum_{j=0}^{n_2-1} F_M(t - t_j^2) D_N(t_0^1 - t_j^2) \eta_0^1 \eta_{t_{j+1}}^2 \\ & + \sum_{j=0}^{n_2-1} F_M(t - t_j^2) D_N(t_n^1 - t_j^2) \eta_{t_n}^1 \eta_{t_{j+1}}^2 \\ & - \sum_{i=0}^{n_1-1} F_M(t - t_0^2) D_N(t_j^1 - t_0^2) \eta_{t_{i+1}}^1 \eta_0^2 \\ & + \sum_{i=0}^{n_1-1} F_M(t - t_n^2) D_N(t_j^1 - t_n^2) \eta_{t_{i+1}}^1 \eta_{t_n}^2 \\ & + \sum_{j=0}^{n_2-1} F_M(t - t_j^2) D_N(t_0^1 - t_j^2) \eta_0^1 \eta_j^2 \\ & - \sum_{j=0}^{n_2-1} F_M(t - t_j^2) D_N(t_n^1 - t_j^2) \eta_{t_n}^1 \eta_j^2 \\ & + \sum_{i=0}^{n_1-1} F_M(t - t_0^2) D_N(t_i^1 - t_0^2) \eta_i^1 \eta_0^2 \\ & - \sum_{i=0}^{n_1-1} F_M(t - t_n^2) D_N(t_i^1 - t_n^2) \eta_i^1 \eta_{t_n}^2. \end{aligned} \quad (A28)$$

$$\begin{aligned} & - \sum_{j=0}^{n_2-1} F_M(t - t_j^2) D_N(t_0^1 - t_j^2) \eta_0^1 \eta_{t_{j+1}}^2 \\ & + \sum_{j=0}^{n_2-1} F_M(t - t_j^2) D_N(t_n^1 - t_j^2) \eta_{t_n}^1 \eta_{t_{j+1}}^2 \\ & - \sum_{i=0}^{n_1-1} F_M(t - t_0^2) D_N(t_j^1 - t_0^2) \eta_{t_{i+1}}^1 \eta_0^2 \\ & + \sum_{i=0}^{n_1-1} F_M(t - t_n^2) D_N(t_j^1 - t_n^2) \eta_{t_{i+1}}^1 \eta_{t_n}^2 \\ & + \sum_{j=0}^{n_2-1} F_M(t - t_j^2) D_N(t_0^1 - t_j^2) \eta_0^1 \eta_j^2 \\ & - \sum_{j=0}^{n_2-1} F_M(t - t_j^2) D_N(t_n^1 - t_j^2) \eta_{t_n}^1 \eta_j^2 \\ & + \sum_{i=0}^{n_1-1} F_M(t - t_0^2) D_N(t_i^1 - t_0^2) \eta_i^1 \eta_0^2 \\ & - \sum_{i=0}^{n_1-1} F_M(t - t_n^2) D_N(t_i^1 - t_n^2) \eta_i^1 \eta_{t_n}^2. \end{aligned} \quad (A29)$$

$$\begin{aligned} & - \sum_{j=0}^{n_2-1} F_M(t - t_j^2) D_N(t_0^1 - t_j^2) \eta_0^1 \eta_{t_{j+1}}^2 \\ & + \sum_{j=0}^{n_2-1} F_M(t - t_j^2) D_N(t_n^1 - t_j^2) \eta_{t_n}^1 \eta_{t_{j+1}}^2 \\ & - \sum_{i=0}^{n_1-1} F_M(t - t_0^2) D_N(t_j^1 - t_0^2) \eta_{t_{i+1}}^1 \eta_0^2 \\ & + \sum_{i=0}^{n_1-1} F_M(t - t_n^2) D_N(t_j^1 - t_n^2) \eta_{t_{i+1}}^1 \eta_{t_n}^2 \\ & + \sum_{j=0}^{n_2-1} F_M(t - t_j^2) D_N(t_0^1 - t_j^2) \eta_0^1 \eta_j^2 \\ & - \sum_{j=0}^{n_2-1} F_M(t - t_j^2) D_N(t_n^1 - t_j^2) \eta_{t_n}^1 \eta_j^2 \\ & + \sum_{i=0}^{n_1-1} F_M(t - t_0^2) D_N(t_i^1 - t_0^2) \eta_i^1 \eta_0^2 \\ & - \sum_{i=0}^{n_1-1} F_M(t - t_n^2) D_N(t_i^1 - t_n^2) \eta_i^1 \eta_{t_n}^2. \end{aligned} \quad (A30)$$

$$\begin{aligned} & - \sum_{j=0}^{n_2-1} F_M(t - t_j^2) D_N(t_0^1 - t_j^2) \eta_0^1 \eta_{t_{j+1}}^2 \\ & + \sum_{j=0}^{n_2-1} F_M(t - t_j^2) D_N(t_n^1 - t_j^2) \eta_{t_n}^1 \eta_{t_{j+1}}^2 \\ & - \sum_{i=0}^{n_1-1} F_M(t - t_0^2) D_N(t_j^1 - t_0^2) \eta_{t_{i+1}}^1 \eta_0^2 \\ & + \sum_{i=0}^{n_1-1} F_M(t - t_n^2) D_N(t_j^1 - t_n^2) \eta_{t_{i+1}}^1 \eta_{t_n}^2 \\ & + \sum_{j=0}^{n_2-1} F_M(t - t_j^2) D_N(t_0^1 - t_j^2) \eta_0^1 \eta_j^2 \\ & - \sum_{j=0}^{n_2-1} F_M(t - t_j^2) D_N(t_n^1 - t_j^2) \eta_{t_n}^1 \eta_j^2 \\ & + \sum_{i=0}^{n_1-1} F_M(t - t_0^2) D_N(t_i^1 - t_0^2) \eta_i^1 \eta_0^2 \\ & - \sum_{i=0}^{n_1-1} F_M(t - t_n^2) D_N(t_i^1 - t_n^2) \eta_i^1 \eta_{t_n}^2. \end{aligned} \quad (A31)$$

We begin by studying the terms (A26) and (A27), that can be written as follows:

$$\begin{aligned} & - \sum_{i=0}^{n_1-1} \sum_{j=0}^{n_2-1} F_M(t - t_j^2) (D_N(t_{i+1}^1 - t_j^2) - D_N(t_i^1 - t_j^2)) \eta_{t_{i+1}}^1 \eta_{t_{j+1}}^2 \\ & + \sum_{i=0}^{n_1-1} \sum_{j=0}^{n_2-1} F_M(t - t_{j+1}^2) (D_N(t_{i+1}^1 - t_{j+1}^2) - D_N(t_i^1 - t_{j+1}^2)) \\ & \times \eta_{t_{i+1}}^1 \eta_{t_{j+1}}^2. \end{aligned} \quad (A32)$$

$$\begin{aligned} & + \sum_{i=0}^{n_1-1} \sum_{j=0}^{n_2-1} F_M(t - t_{j+1}^2) (D_N(t_{i+1}^1 - t_{j+1}^2) - D_N(t_i^1 - t_{j+1}^2)) \\ & \times \eta_{t_{i+1}}^1 \eta_{t_{j+1}}^2. \end{aligned} \quad (A33)$$

Using the Taylor formula, we express (A32) and (A33) as

$$\begin{aligned} & \sum_{i=0}^{n_1-1} \sum_{j=0}^{n_2-1} \left[F_M(t - t_{j+1}^2) D'_N(t_i^1 - t_{j+1}^2) \rho(n_1) \right. \\ & \quad \left. - F_M(t - t_j^2) D'_N(t_i^1 - t_j^2) \rho(n_1) \right] \eta_{t_{i+1}}^1 \eta_{t_{j+1}}^2 + o(\rho(n_1)) \\ &= \sum_{i=0}^{n_1-1} \sum_{j=0}^{n_2-1} \left[F'_M(t - t_j^2) D'_N(t_i^1 - t_j^2) + F_M(t - t_j^2) D''_N(t_i^1 - t_j^2) \right] \\ & \quad \times \rho(n_1) \rho(n_2) \eta_{t_{i+1}}^1 \eta_{t_{j+1}}^2 + o(\rho(n_1) \rho(n_2)). \end{aligned}$$

We study the contribution to the asymptotic variance of these two terms:

$$\begin{aligned} E_{31}(t) &:= \frac{1}{2\pi} \sum_{i=0}^{n_1-1} \sum_{j=0}^{n_2-1} F'_M(t - t_j^2) D'_N(t_i^1 - t_j^2) \\ &\quad \times \rho(n_1) \rho(n_2) \eta_{i+1}^1 \eta_{j+1}^2, \\ E_{32}(t) &:= \frac{1}{2\pi} \sum_{i=0}^{n_1-1} \sum_{j=0}^{n_2-1} F_M(t - t_j^2) D''_N(t_i^1 - t_j^2) \rho(n_1) \rho(n_2) \\ &\quad \times \eta_{i+1}^1 \eta_{j+1}^2. \end{aligned}$$

Consider $E_{31}(t)$. It holds:

$$\begin{aligned} &E[(\rho(n)^{-1/4} M^{-1/2} \frac{1}{2\pi} \sum_{i=0}^{n_1-1} \sum_{j=0}^{n_2-1} F'_M(t - t_j^2) D'_N(t_i^1 - t_j^2) \rho(n_1) \\ &\quad \times \rho(n_2) \eta_{i+1}^1 \eta_{j+1}^2)^2] \\ &= \rho(n)^{-1/2} M^{-1} \frac{1}{(2\pi)^2} \sum_{i=0}^{n_1-1} \sum_{j=0}^{n_2-1} |F'_M(t - t_j^2)|^2 |D'_N(t_i^1 - t_j^2)|^2 \\ &\quad \times \rho(n_1)^2 \rho(n_2)^2 E[(\eta_{i+1}^1)^2 (\eta_{j+1}^2)^2] \\ &\quad + \rho(n)^{-1/2} M^{-1} \frac{1}{(2\pi)^2} \sum_{i=0}^{n_1-1} \sum_{i' \neq i}^{n_1-1} \sum_{j=0}^{n_2-1} \sum_{j' \neq j}^{n_2-1} F'_M(t - t_j^2) \\ &\quad \times F'_M(t - t_{j'}^2) D'_N(t_i^1 - t_{j'}^2) D'_N(t_{i'}^1 - t_j^2) \rho(n_1)^2 \rho(n_2)^2 \\ &\quad \times E[\eta_{i+1}^1 \eta_{i'+1}^1 \eta_{j+1}^2 \eta_{j'+1}^2]. \end{aligned} \quad (A34)$$

First, we study the term (A34), which, adopting the continuous-time notation and thanks to the independency assumption between η^1 and η^2 , is equal to:

$$\begin{aligned} &\rho(n)^{-1/2} M^{-1} \frac{1}{(2\pi)^2} \int_0^{2\pi} |F'_M(t - \varphi_n^2(v))|^2 \\ &\quad \times \int_0^{2\pi} |D'_N(\varphi_n^1(u) - \varphi_n^2(v))|^2 du dv \rho(n_1) \rho(n_2) \xi_1 \xi_2 \end{aligned}$$

that, in virtue of Lemmas A.5 and A.7, has order

$$\rho(n)^{-1/2} M^{-1} M^3 N \rho(n_1) \rho(n_2) = o(1).$$

Consider now (A35). For the assumption on noise, η_i^1 is independent of η_j^2 for any i, j ; moreover, η_i^1 is independent of η_{i+1}^1 , and the same holds for η_j^2 and η_{j+1}^2 . Thus $E[\eta_{i+1}^1 \eta_{i'+1}^1 \eta_{j+1}^2 \eta_{j'+1}^2] = E[\eta_{i+1}^1] E[\eta_{i'+1}^1] E[\eta_{j+1}^2] E[\eta_{j'+1}^2] = 0$, for the assumption on the moments of the noise, so (A35) is equal to 0.

Now we study the contribution to the asymptotic variance from $E_{32}(t)$. It holds:

$$\begin{aligned} &E[(\rho(n)^{-1/4} M^{-1/2} \frac{1}{2\pi} \sum_{i=0}^{n_1-1} \sum_{j=0}^{n_2-1} F_M(t - t_j^2) D''_N(t_i^1 - t_j^2) \\ &\quad \times \rho(n_1) \rho(n_2) \eta_{i+1}^1 \eta_{j+1}^2)^2] \\ &= \rho(n)^{-1/2} M^{-1} \frac{1}{(2\pi)^2} \sum_{i=0}^{n_1-1} \sum_{j=0}^{n_2-1} F_M^2(t - t_j^2) |D''_N(t_i^1 - t_j^2)|^2 \\ &\quad \times \rho(n_1)^2 \rho(n_2)^2 E[(\eta_{i+1}^1)^2 (\eta_{j+1}^2)^2] \end{aligned} \quad (A36)$$

$$\begin{aligned} &+ \rho(n)^{-1/2} M^{-1} \frac{1}{(2\pi)^2} \sum_{i=0}^{n_1-1} \sum_{i' \neq i}^{n_1-1} \sum_{j=0}^{n_2-1} \sum_{j' \neq j}^{n_2-1} F_M(t - t_j^2) \\ &\quad \times D''_N(t_i^1 - t_{j'}^2) F_M(t - t_{j'}^2) D''_N(t_{i'}^1 - t_j^2) \rho(n_1)^2 \rho(n_2)^2 \\ &\quad \times E[\eta_i^1 \eta_{i'+1}^1 \eta_{j+1}^2 \eta_{j'+1}^2]. \end{aligned} \quad (A37)$$

Term (A37) is equal to 0, analogously to term (A35). Consider now (A36); using the continuous-time notation, it is equal to

$$\begin{aligned} &\rho(n)^{-1/2} M^{-1} \frac{1}{(2\pi)^2} \int_0^{2\pi} F_M^2(t - \varphi_n^2(v)) \\ &\quad \times \int_0^{2\pi} |D''_N(\varphi_n^1(u) - \varphi_n^2(v))|^2 du dv \rho(n_1) \rho(n_2) \xi_1 \xi_2 \end{aligned} \quad (A38)$$

thanks to Lemmas A.3 and A.7, it is equal to:

$$\frac{1}{15c} \xi_1 \xi_2 N^4 s_1 s_2 \rho(n)^2 \rightarrow \frac{1}{15} c^3 s_1 s_2 \xi_1 \xi_2.$$

It remains to prove the asymptotic negligibility of the end points' contribution. The terms are grouped as follows:

$$- \sum_{j=0}^{n_2-1} F_M(t - t_j^2) D_N(t_j^2) \eta_0^1 \eta_{j+1}^2 + \sum_{j=0}^{n_2-1} F_M(t - t_j^2) D_N(t_j^2) \eta_0^1 \eta_j^2 \quad (A39)$$

$$\begin{aligned} &+ \sum_{j=0}^{n_2-1} F_M(t - t_j^2) D_N(t_j^2) \eta_{n_1}^1 \eta_{j+1}^2 - \sum_{j=0}^{n_2-1} F_M(t - t_j^2) \\ &\quad \times D_N(t_j^2) \eta_{n_1}^1 \eta_j^2 \end{aligned} \quad (A40)$$

$$- \sum_{i=0}^{n_1-1} F_M(t) D_N(t_i^1) \eta_{i+1}^1 \eta_0^2 + \sum_{i=0}^{n_1-1} F_M(t) D_N(t_i^1) \eta_i^1 \eta_0^2 \quad (A41)$$

$$+ \sum_{i=0}^{n_1-1} F_M(t) D_N(t_i^1) \eta_{i+1}^1 \eta_{n_2}^2 - \sum_{i=0}^{n_1-1} F_M(t) D_N(t_i^1) \eta_i^1 \eta_{n_2}^2. \quad (A42)$$

We first consider (A41). It can be written as

$$\begin{aligned} &- \sum_{i=0}^{n_1-1} F_M(t) D_N(t_i^1) \eta_{i+1}^1 \eta_0^2 + \sum_{i=0}^{n_1-1} F_M(t) D_N(t_{i+1}^1) \eta_{i+1}^1 \eta_0^2 \\ &\quad + F_M(t) D_N(t_0^1) \eta_0^1 \eta_0^2 - F_M(t) D_N(t_{n_1}^1) \eta_{n_1}^1 \eta_0^2 \\ &= F_M(t) \sum_{i=0}^{n_1-1} (D_N(t_{i+1}^1) - D_N(t_i^1)) \eta_{i+1}^1 \eta_0^2 \\ &\quad + F_M(t) \eta_0^1 \eta_0^2 - F_M(t) \eta_{n_1}^1 \eta_0^2, \end{aligned}$$

where we used that $D_N(t_0^1) = D_N(0) = 1 = D_N(2\pi) = D_N(t_{n_1}^1)$. Notice that, for the i.i.d property of the noise, it holds:

$$\begin{aligned} & \rho(n)^{-1/2} M^{-1} E \left[\left(F_M(t) \sum_{i=0}^{n_1-1} D'_N(t_i^1) \rho(n_1) \eta_{i+1}^1 \eta_0^2 \right. \right. \\ & \quad \left. \left. + F_M(t) \eta_0^1 \eta_0^2 - F_M(t) \eta_{n_1}^1 \eta_0^2 \right)^2 \right] \\ &= \rho(n)^{-1/2} M^{-1} \\ & \quad \times \left(F_M^2(t) \sum_{i=0}^{n_1-1} |D'_N(t_i^1)|^2 \rho(n_1)^2 \xi_1 \xi_2 + 2F_M^2(t) \xi_1 \xi_2 \right). \end{aligned}$$

Finally, observe that

$$\begin{aligned} & \rho(n)^{-1/2} M^{-1} \left(F_M^2(t) \rho(n_1) \int_0^{2\pi} |D'_N(\varphi_n^1(s))|^2 ds \xi_1 \xi_2 \right. \\ & \quad \left. + F_M^2(t) \xi_1 \xi_2 \right) = O(\rho(n)^{1/2} M^{-3} N) + O(\rho(n)^{-1/2} M^{-3}) = o(1), \end{aligned} \quad (A43)$$

where we used the fact that, for any $\varepsilon > 0$ and $\varepsilon \leq t \leq 2\pi - \varepsilon$, $F_M(t) \leq M^{-1} (\sin(\varepsilon/2))^{-2}$, and Lemma A.6.

Consider now (A39). With a similar decomposition as for (A41), it can be written as

$$\begin{aligned} & \sum_{j=0}^{n_2-1} \left[\left(F'_M(t - t_j^2) D_N(t_j^2) + F_M(t - t_j^2) D'_N(t_j^2) \right) \rho(n_2) \right. \\ & \quad \left. + o(\rho(n_2)) \eta_0^1 \eta_{j+1}^2 + F_M(t) \eta_0^1 \eta_0^2 - F_M(t) \eta_0^1 \eta_{n_2}^2 \right], \end{aligned}$$

where we used that $D_N(t_0^2) = D_N(0) = 1 = D_N(2\pi) = D_N(t_{n_2}^2)$. Then, for the i.i.d property of the noise, it holds:

$$\begin{aligned} & \rho(n)^{-1/2} M^{-1} E \left[\left(\sum_{j=0}^{n_2-1} \left(F'_M(t - t_j^2) D_N(t_j^2) + F_M(t - t_j^2) D'_N(t_j^2) \right) \right. \right. \\ & \quad \left. \left. \times \rho(n_2) \eta_0^1 \eta_{j+1}^2 + F_M(t) \eta_0^1 \eta_0^2 - F_M(t) \eta_0^1 \eta_{n_2}^2 \right)^2 \right] \\ &= \rho(n)^{-1/2} M^{-1} \left(\sum_{j=0}^{n_2-1} \left(|F'_M(t - t_j^2)|^2 D_N^2(t_j^2) \right. \right. \\ & \quad \left. \left. + F_M^2(t - t_j^2) |D'_N(t_j^2)|^2 \right) \rho(n_2)^2 \xi_1 \xi_2 + 2F_M^2(t) \xi_1 \xi_2 \right). \end{aligned}$$

For the first term, we have:

$$\begin{aligned} & \rho(n)^{-1/2} M^{-1} \rho(n_2) \int_0^{2\pi} |F'_M(t - \varphi_n^2(s))|^2 D_N^2(\varphi_n^2(s)) ds \xi_1 \xi_2 \\ & \sim \rho(n)^{-1/2} M^{-1} \rho(n_2) M^3 D_N^2(t), \end{aligned}$$

where we have used Lemma A.5. Moreover, using that for $t \in (0, 2\pi)$, then $D_N^2(t) = O(N^{-1})$, this term is

$$O(\rho(n)^{1/2} M^2 N^{-1}) = o(1).$$

For the second term, we study:

$$\rho(n)^{-1/2} M^{-1} \rho(n_2) \int_0^{2\pi} F_M^2(t - \varphi_n^2(s)) |D'_N(\varphi_n^2(s))|^2 ds \xi_1 \xi_2.$$

Using Lemmas A.3 and A.7, this gives, for $t \neq 0, 2\pi$, that

$$\rho(n)^{-1/2} M^{-1} \rho(n_2) M |D'_N(t)|^2 = O(\rho(n)^{1/2} N) o(1) = o(1).$$

Finally, consider the remaining term, which is equal to

$$\begin{aligned} & 2 \rho(n)^{-1/2} M^{-1} F_M^2(t) \xi_1 \xi_2 \\ &= O(\rho(n)^{-1/2} M^{-1} M^{-2}) = O(\rho(n)^{1/4}). \end{aligned}$$

The terms (A40) and (A42) are analogous.

A.1.3. Mixed noise-price component. Consider now the asymptotic variance contribution from the terms (A5) and (A6), which are analogous. Thus we study only $\langle E_1(t), E_1(t) \rangle$. We observe that the term $E_1(t)$ can be written as:

$$\begin{aligned} & \sum_{i=0}^{n_1-1} \sum_{j=0}^{n_2-1} F_M(t - t_j^2) D_N(t_i^1 - t_j^2) \delta_i(p^1) \eta_{j+1}^2 \\ & - \sum_{i=0}^{n_1-1} \sum_{j=0}^{n_2-1} F_M(t - t_{j+1}^2) D_N(t_i^1 - t_{j+1}^2) \delta_i(p^1) \eta_{j+1}^2 \end{aligned} \quad (A44)$$

plus the remaining end-terms equal to

$$\begin{aligned} & - \sum_{i=0}^{n_1-1} F_M(t - t_0^2) D_N(t_i^1 - t_0^2) \delta_i(p^1) \eta_0^2 \\ & + \sum_{i=0}^{n_1-1} F_M(t - t_{n_2}^2) D_N(t_i^1 - t_{n_2}^2) \delta_i(p^1) \eta_{n_2}^2. \end{aligned} \quad (A45)$$

Consider (A44). It is equal to

$$\sum_{i=0}^{n_1-1} \sum_{j=0}^{n_2-1} F'_M(t - t_j^2) D_N(t_i^1 - t_j^2) \delta_i(p^1) \eta_{j+1}^2 \rho(n_2) + o(\rho(n_2)) \quad (A46)$$

$$+ \sum_{i=0}^{n_1-1} \sum_{j=0}^{n_2-1} F_M(t - t_j^2) D'_N(t_i^1 - t_j^2) \delta_i(p^1) \eta_{j+1}^2 \rho(n_2) + o(\rho(n_2)). \quad (A47)$$

Thus we first consider the term (A46). For the i.i.d. property for the noise and the independence between the noise and price process, considering the conditional expectation with respect to the price process p , it holds:

$$\begin{aligned} & E \left[\left(\rho(n)^{-1/4} M^{-1/2} \frac{1}{2\pi} \sum_{i=0}^{n_1-1} \sum_{j=0}^{n_2-1} F'_M(t - t_j^2) \right. \right. \\ & \quad \left. \left. \times D_N(t_i^1 - t_j^2) \rho(n_2) \delta_i(p^1) \eta_{j+1}^2 \right)^2 \right]. \end{aligned}$$

It is enough to study the rate of the quadratic part:

$$\begin{aligned} & \rho(n)^{-1/2} M^{-1} \frac{1}{(2\pi)^2} \sum_{i=0}^{n_1-1} \sum_{j=0}^{n_2-1} |F'_M(t - t_j^2)|^2 D_N^2(t_i^1 - t_j^2) \delta_i^2(p^1) \\ & \times E \left[(\eta_{j+1}^2)^2 \right] \rho(n_2)^2. \end{aligned} \quad (A48)$$

Moving to the continuous-time notation, we get:

$$\begin{aligned} & \rho(n)^{-1/2} M^{-1} \frac{1}{(2\pi)^2} \int_0^{2\pi} |F'_M(t - \varphi_n^2(v))|^2 \\ & \times \int_0^{2\pi} D_N^2(\varphi_n^1(u) - \varphi_n^2(v)) \Sigma^{1,1}(u) du dv \frac{s_2^2}{s_1} \rho(n) \xi_2, \end{aligned}$$

which, thanks to Lemmas A.5 and A.6, is equal to

$$\begin{aligned} & \rho(n)^{-1/2} M^{-1} \frac{1}{(2\pi)^2} M^3 \frac{2\pi}{15} \frac{1}{N} \pi \Sigma^{1,1}(t) \frac{s_2^2}{s_1} \rho(n) \xi_2 \sim C \rho(n)^{1/2} \frac{M^2}{N} \\ &= o_p(1). \end{aligned}$$

Then consider the term (A47). It holds:

$$\begin{aligned}
& \left(\rho(n)^{-1/4} M^{-1/2} \frac{1}{2\pi} \sum_{i=0}^{n_1-1} \sum_{j=0}^{n_2-1} F_M(t - t_j^2) D'_N(t_i^1 - t_j^2) \right. \\
& \quad \left. \times \rho(n_2) \delta_i(p^1) \eta_{j+1}^2 \right)^2 \\
&= \rho(n)^{-1/2} M^{-1} \frac{1}{(2\pi)^2} \sum_{i=0}^{n_1-1} \sum_{j=0}^{n_2-1} F_M^2(t - t_j^2) |D'_N(t_i^1 - t_j^2)|^2 \\
& \quad \times \rho(n_2)^2 \delta_i^2(p^1) (\eta_{j+1}^2)^2 \quad (\text{A49}) \\
&+ \rho(n)^{-1/2} M^{-1} \frac{1}{(2\pi)^2} \sum_{i=0}^{n_1-1} \sum_{\substack{i'=0 \\ i' \neq i}}^{n_1-1} \sum_{j=0}^{n_2-1} \sum_{\substack{j'=0 \\ j' \neq j}}^{n_2-1} F_M^2(t - t_j^2) \\
& \quad \times |D'_N(t_i^1 - t_j^2)|^2 \rho(n_2)^2 \delta_i(p^1) \delta_{i'}(p^1) \eta_j^2 \eta_{j'+1}^2. \quad (\text{A50})
\end{aligned}$$

Consider (A49). The conditional expectation with respect to p gives

$$\frac{1}{c} \frac{N}{M} \rho(n_2)^2 \frac{1}{(2\pi)^2} \sum_{i=0}^{n_1-1} \sum_{j=0}^{n_2-1} F_M^2(t - t_j^2) |D'_N(t_i^1 - t_j^2)|^2 \delta_i^2(p^1) \xi_2,$$

and, taking the continuous-time notation, it holds that

$$\begin{aligned}
& \frac{1}{c} \frac{1}{(2\pi)^2} N^2 \rho(n_2) \left(\frac{1}{M} \int_0^{2\pi} F_M^2(t - \varphi_n^2(s)) \frac{1}{N} \right. \\
& \quad \left. \times \int_0^{2\pi} |D'_N(\varphi_n^1(u) - \varphi_n^2(v))|^2 \Sigma^{1,1}(u) du dv + o_p(1) \right) \xi_2 \\
&= \frac{1}{c} \frac{1}{(2\pi)^2} N^2 \rho(n_2) \frac{4\pi}{3} \frac{\pi}{3} \sigma_1^2(t) \xi_2 \rightarrow cs_2 \frac{1}{9} \Sigma^{1,1}(t) \xi_2
\end{aligned}$$

in virtue of Lemmas A.3 and A.7.

As for the term (A50), this gives zero contribution for the i.i.d property of the noise process and the fact that it is zero-mean.

Consider now the end-points term in (A45) and prove their asymptotic negligibility. Consider the first addend:

$$\begin{aligned}
& \left(n^{1/4} M^{-1/2} \sum_{i=0}^{n_1-1} F_M(t) D_N(t_i^1) \delta_i(p^1) \eta_0^2 \right)^2 \\
&= n^{1/2} M^{-1} \sum_{i=0}^{n_1-1} F_M^2(t) D_N^2(t_i^1) (\delta_i(p^1))^2 (\eta_0^2)^2 \quad (\text{A51}) \\
&+ n^{1/2} M^{-1} 2 \sum_i \sum_{i' < i} F_M^2(t) D_N(t_i^1) D_N(t_{i'}^1) \delta_i(p^1) \delta_{i'}(p^1) (\eta_0^2)^2. \quad (\text{A52})
\end{aligned}$$

The conditional expectation with respect to p of (A51) gives

$$n^{1/2} M^{-1} F_M^2(t) \left(\int_0^{2\pi} D_N^2(\varphi_n^1(u)) \Sigma^{1,1}(u) du + o_p(1) \right) \xi_2.$$

Noting that, for any $\varepsilon > 0$ and $\varepsilon \leq t \leq 2\pi - \varepsilon$, $F_M(t) \leq M^{-1} (\sin(\varepsilon/2))^{-2}$, and using Lemma A.6, then (A51) has order $n^{1/2} M^{-3} N^{-1} = o(1)$.

Consider now (A52). Given the independence between the noise and price processes, we get:

$$\begin{aligned}
& n^{1/2} M^{-1} 2 F_M^2(t) \int_0^{2\pi} \int_0^s D_N(\varphi_n^1(u)) \sigma_1^*(u) dW_u D_N(\varphi_n^1(s)) \\
& \quad \times \sigma_1^*(s) dW_s \xi_2.
\end{aligned}$$

It holds:

$$\begin{aligned}
& E \left[\left(\int_0^{2\pi} \int_0^s D_N(\varphi_n^1(u)) \sigma_1^*(u) dW_u D_N(\varphi_n^1(s)) \sigma_1^*(s) dW_s \right)^2 \right]^{1/2} \\
& \leq C \int_0^{2\pi} D_N^2(u) du = C \frac{2\pi}{2N+1}.
\end{aligned}$$

Therefore (A52) has order $O_p(n^{1/2} M^{-2} N^{-1}) = o_p(1)$.

For the second addend in (A45) it is enough to observe that $t_n^2 = 2\pi$ and use the periodicity of the kernels and the properties of the noise process.

The contribution of the term $E_2(t)$ in (A6) is analogous, giving

$$cs_1 \frac{1}{9} \Sigma^{2,2}(t) \xi_1.$$

A.1.4. Asymptotic orthogonality. The last step needed to ensure the stable convergence requires proving the following convergence in probability:

$$\begin{aligned}
& \langle \rho(n)^{-1/4} M^{-1/2} (F_1(t) + F_2(t) + G(t)), W^r \rangle_{2\pi} \\
& \rightarrow 0, \quad \forall r = 1, \dots, K.
\end{aligned}$$

We start considering $F_1(t)$. For all $r = 1, \dots, K$ we have:

$$\begin{aligned}
& \left\langle \rho(n)^{-1/4} M^{-1/2} F_1(t), W^r \right\rangle \\
&= \rho(n)^{-1/4} M^{-1/2} \frac{1}{2\pi} \int_0^{2\pi} F_M(t - \varphi_n^2(v)) \\
& \quad \times \int_0^v D_N(\varphi_n^1(u) - \varphi_n^2(v)) \sigma_1^*(u) dW_u \sigma_{2,r}(v) dv
\end{aligned}$$

taking the L^1 -norm:

$$\begin{aligned}
& E \left[\left| \rho(n)^{-1/4} M^{-1/2} \frac{1}{2\pi} \int_0^{2\pi} F_M(t - \varphi_n^2(v)) \right. \right. \\
& \quad \left. \times \int_0^v D_N(\varphi_n^1(u) - \varphi_n^2(v)) \sigma_1^*(u) dW_u \sigma_{2,r}(v) dv \right| \Big] \\
& \leq \rho(n)^{-1/4} M^{-1/2} \frac{1}{2\pi} \|\sigma_{2,r}\|_\infty \int_0^{2\pi} F_M(t - \varphi_n^2(v)) \\
& \quad \times E \left[\left(\int_0^v D_N(\varphi_n^1(u) - \varphi_n^2(v)) \sigma_1^*(u) dW_u \right)^2 \right]^{1/2} dv. \quad (\text{A53})
\end{aligned}$$

We have that

$$\begin{aligned}
& E \left[\left(\int_0^v D_N(\varphi_n^1(u) - \varphi_n^2(v)) \sigma_1^*(u) dW_u \right)^2 \right] \\
& \leq \|\Sigma^{1,1}\|_\infty \int_0^v D_N^2(\varphi_n^1(u) - \varphi_n^2(v)) du.
\end{aligned}$$

Therefore, by Lemmas A.6 and A.3(i), the term (A53) is less or equal to

$$\begin{aligned}
& C \rho(n)^{-1/4} M^{-1/2} N^{-1/2} \int_0^{2\pi} F_M(t - \varphi_n^2(v)) \\
& \quad \times \left(N \int_0^v D_N^2(\varphi_n^1(u) - \varphi_n^2(v)) du \right)^{1/2} dv = O(M^{-1/2}).
\end{aligned}$$

The term $F_2(t)$ is analogous to $F_1(t)$. Considering $G(t)$, it reduces to two analogous terms given by (A17) and (A18). Focusing on (A17)

we get:

$$\begin{aligned} & \rho(n)^{-1/4} M^{-1/2} \frac{1}{2\pi} \sum_{0 < |k| \leq M} \left(1 - \frac{|k|}{M+1}\right)^2 \\ & \times \frac{1}{k^2} \int_t^{t+\frac{2\pi}{M+1}} \gamma_{1,2}(s) ds \phi^{1,2,r} \end{aligned}$$

with $\phi^{1,2,r} = \langle Z^{1,2}, W^r \rangle$ with $Z^{1,2}$ being the Brownian Motion driving the process $\Sigma^{1,2}(t)$.

$$\begin{aligned} & E \left[\left| \rho(n)^{-1/4} M^{-1/2} \frac{1}{2\pi} \sum_{0 < |k| \leq M} \left(1 - \frac{|k|}{M+1}\right) \frac{i}{k} \right. \right. \\ & \quad \times \left. \left. \int_{t-\frac{2\pi}{M+1}}^t \gamma_{1,2}(s) ds \right| \right] \\ & \leq C \rho(n)^{-1/4} M^{-1/2} \frac{1}{2\pi} \sum_{0 < |k| \leq M} \left(1 - \frac{|k|}{M+1}\right) \frac{1}{|k|} \frac{2\pi}{M+1} \\ & \quad \times \left(\gamma_{1,2} \left(t - \frac{2\pi}{M+1} \right) + O_p \left(\frac{1}{M+1} \right) \right) \\ & \leq C \rho(n)^{-1/4} M^{-1/2} \frac{1}{M+1} \sum_{k=1}^M \left(1 - \frac{k}{M+1}\right) \frac{1}{k} \\ & \sim \frac{1}{\sqrt{M}} \left(\ln M + \bar{\gamma} + O \left(\frac{1}{M} \right) \right) \rightarrow 0. \end{aligned}$$

Finally, with a conditional expectation argument and the fact that at least one of the terms in the product has expectation equal to zero, it is easily seen that the covariance of $F_1(t) + F_2(t) + G(t)$ and $E_1(t) + E_2(t) + E_3(t)$ is asymptotically zero.

A.2. Proof of Proposition 3.3

Proof The consistency of the estimator $\tilde{\Sigma}_{n,N,M}^{1,2}(t)$ follows from Theorem 3.1. Further, the consistency of the spot variance estimator $\tilde{\Sigma}_{n,N,M}^{1,2}(t)$ is shown in Mancino et al. (2024), and its positivity has been proved in Malliavin and Mancino (2009, Remark 2.3). Then, the Continuous Mapping Theorem (Mann and Wald 1943) ensures the consistency of the ratio $\frac{\tilde{\Sigma}_{n,N,M}^{1,2}(t)}{\tilde{\Sigma}_{n,N,M}^{2,2}(t)}$. ■

Appendix B. Auxiliary results on kernels

This Appendix resumes some results about the rescaled Dirichlet kernel, defined as

$$D_N(x) := \frac{1}{2N+1} \sum_{|k| \leq N} e^{ikx} = \frac{1}{2N+1} \frac{\sin((2N+1)x/2)}{\sin(x/2)} \quad (\text{A54})$$

and the Fejér kernel, defined as

$$\begin{aligned} F_M(x) &:= \sum_{|k| \leq M} \left(1 - \frac{|k|}{M+1}\right) e^{ikx} \\ &= \frac{1}{M+1} \left(\frac{\sin((M+1)x/2)}{\sin(x/2)} \right)^2. \end{aligned} \quad (\text{A55})$$

LEMMA A.3 (i) For any M ,

$$\int_{-\pi}^{\pi} F_M(x) dx = 2\pi.$$

Moreover, under the assumption $\lim_{n,M \rightarrow \infty} M^\tau \rho(n) = a$, for a constant $a > 0$ and for some $\tau > 1$, it holds that

$$\lim_{n,M \rightarrow \infty} \int_{-\pi}^{\pi} F_M(\varphi_n(x)) dx = 2\pi$$

and, in particular,

$$\left| \int_{-\pi}^{\pi} F_M(\varphi_n(x)) dx - \int_{-\pi}^{\pi} F_M(x) dx \right| \leq C \frac{M}{n}.$$

(ii) For any $M \geq 1$,

$$\int_{\frac{2\pi}{M+1}}^{\pi} F_M(x) dx \leq \frac{C}{M}.$$

(iii) Under the assumption $\lim_{n,M \rightarrow \infty} M^\tau \rho(n) = a$, for a constant $a > 0$ and for some $\tau > 1$, it holds that

$$\begin{aligned} \lim_{M \rightarrow \infty} \int_{-\pi}^{\pi} \frac{1}{M} F_M^2(x) dx &= \lim_{M,n \rightarrow \infty} \int_{-\pi}^{\pi} \frac{1}{M} F_M^2(\varphi_n(x)) dx \\ &= \frac{4\pi}{3}. \end{aligned} \quad (\text{A56})$$

(iv) If $\lim_{n,M \rightarrow \infty} M^\tau \rho(n) = a$, for a constant $a > 0$ and for some $\tau > 1$, and f is Hölder continuous with parameter $\alpha \in (0, 1]$, then

$$\begin{aligned} \lim_{n,M \rightarrow \infty} \frac{1}{M} \int_{-\pi}^{\pi} F_M^2(x - \varphi_n(y)) f(y) dy \\ = \lim_{M \rightarrow \infty} \int_{-\pi}^{\pi} \frac{1}{M} F_M^2(x - y) f(y) dy = \frac{4\pi}{3} f(x). \end{aligned} \quad (\text{A57})$$

Proof See Cuchiero and Teichmann (2015), Lemma 5.1. ■

REMARK A.4 As observed in Cuchiero and Teichmann (2015), Remark 5.2, all the above results are similarly true on $[0, T]$.

LEMMA A.5 Suppose that $\lim_{n,M \rightarrow \infty} M^\tau \rho(n) = a$, for some constant $a > 0$ and $\tau > 1$. Then, it holds that:

$$\begin{aligned} (i) \quad \lim_{n,M \rightarrow \infty} \int_{-\pi}^{\pi} \frac{1}{M^3} |F'_M(\varphi_n(x))|^2 dx \\ = \lim_{M \rightarrow \infty} \int_{-\pi}^{\pi} \frac{1}{M^3} |F'_M(x)|^2 dx = \frac{2}{15} \pi; \end{aligned} \quad (\text{A58})$$

$$\begin{aligned} (ii) \quad \lim_{n,M \rightarrow \infty} \int_{-\pi}^{\pi} \frac{1}{M^5} |F''_M(\varphi_n(x))|^2 dx \\ = \lim_{M \rightarrow \infty} \int_{-\pi}^{\pi} \frac{1}{M^5} |F''_M(x)|^2 dx = \frac{4}{105} \pi. \end{aligned} \quad (\text{A59})$$

Proof See Toscano et al. (2024), Lemma 8.1. ■

LEMMA A.6 Assume that hypothesis (S) holds. If $\lim_{N,n \rightarrow \infty} N^\tau \rho(n) = c$, for a constant $c > 0$ and some $\tau > 1$, then

$$\begin{aligned} \lim_{n,N \rightarrow \infty} N \int_0^x D_N^2(\varphi_n^1(x) - \varphi_n^2(y)) dy &= \lim_{N \rightarrow \infty} N \int_0^x D_N^2(x - y) dy \\ &= \frac{\pi}{2}. \end{aligned}$$

Proof See Cuchiero and Teichmann (2015), Lemma 5.1, using the fact that $D_N^2(x) = (2N+1)^{-1} F_{2N}(x)$. ■

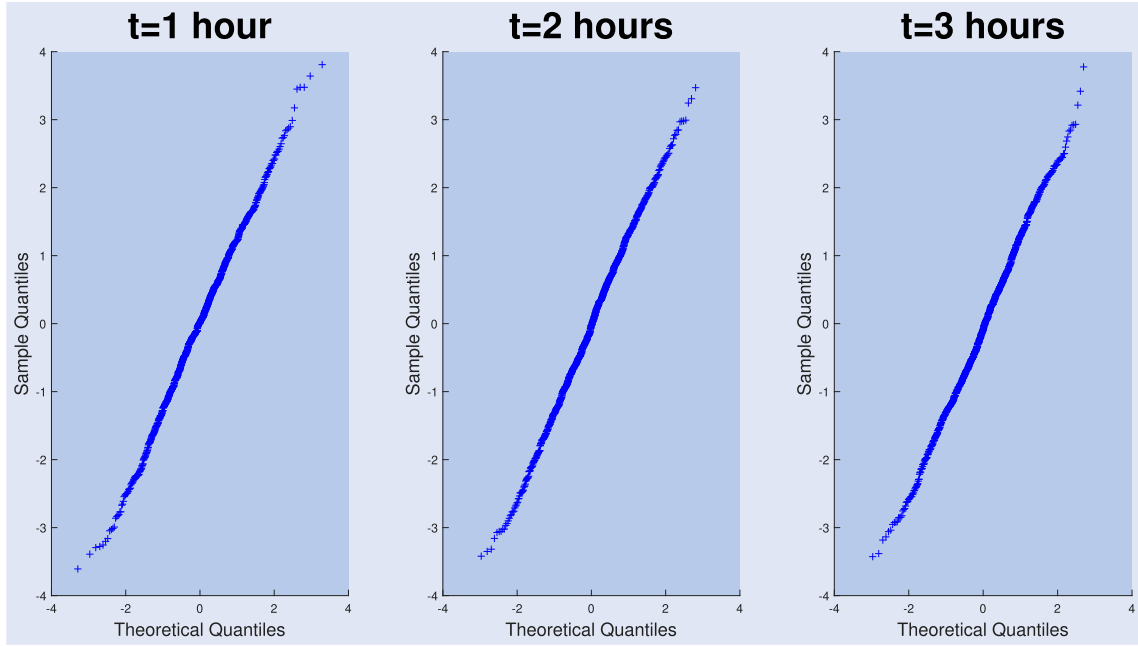


Figure A1. Quantile–quantile plots for the standardized estimation error $\mathcal{E}_{n,N,M}(t)$ for different values of t in the baseline scenario of Section 4.1.2 when $\nu_1 = 1.5$.

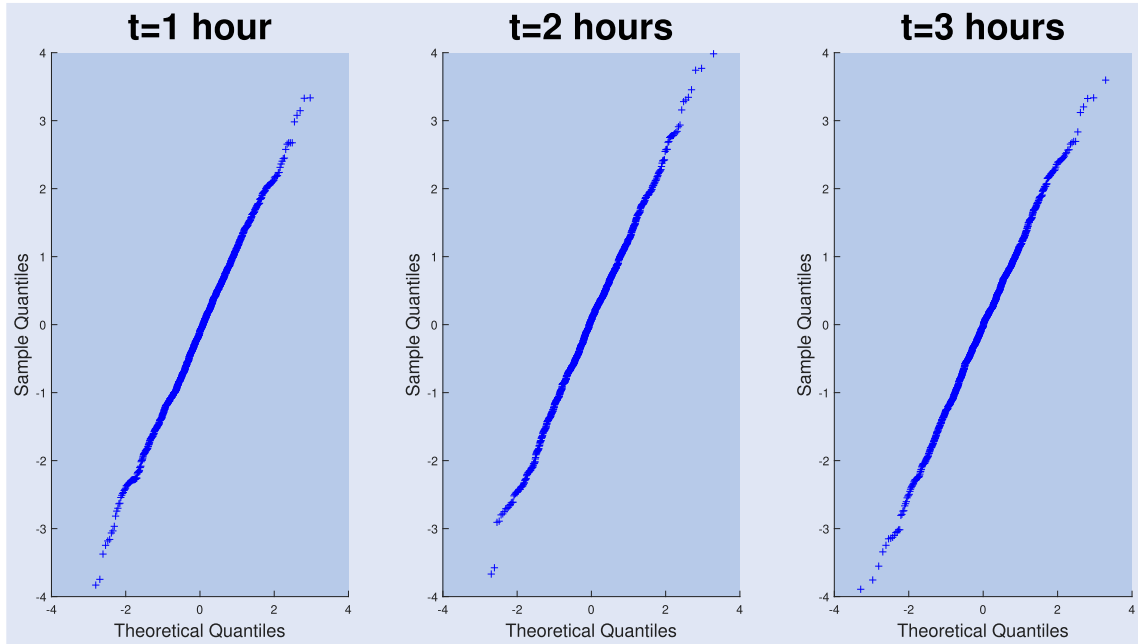


Figure A2. Quantile–quantile plots for the standardized estimation error $\mathcal{E}_{n,N,M}(t)$ for different values of t in the baseline scenario of Section 4.1.2 when $\nu_1 = 2.5$.

LEMMA A.7 Assume that hypothesis (S) holds. Under the condition $\lim_{N,n \rightarrow \infty} N^\tau \rho(n) = c$, for a constant $c > 0$ and some $\tau > 1$, and a Lipschitz function f , it holds that:

$$\begin{aligned} \text{(i)} \quad & \lim_{n,N \rightarrow \infty} \frac{1}{N} \int_0^{2\pi} |D'_N(\varphi_n^1(x) - \varphi_n^2(y))|^2 f(y) dy \\ &= \lim_{N \rightarrow \infty} \frac{1}{N} \int_0^{2\pi} |D'_N(x - y)|^2 f(y) dy = \frac{\pi}{3} f(x); \\ \text{(ii)} \quad & \lim_{n,N \rightarrow \infty} \frac{1}{N^3} \int_0^{2\pi} |D''_N(\varphi_n^1(x) - \varphi_n^2(y))|^2 f(y) dy \\ &= \lim_{N \rightarrow \infty} \frac{1}{N^3} \int_0^{2\pi} |D''_N(x - y)|^2 f(y) dy = \frac{\pi}{5} f(x). \end{aligned}$$

Proof For (i) see Mancino and Toscano (2022), Lemma A.3. As for (ii), it holds that:

$$\begin{aligned} & \frac{1}{N^3} \int_0^{2\pi} |D''_N(x)|^2 dx \\ &= \frac{1}{N^3} \frac{2\pi}{(2N+1)^2} \sum_{|k| \leq N} k^4 \\ &= \frac{2\pi}{15} \frac{N(N+1)(3N^2+3N-1)}{N^3(2N+1)} \rightarrow \frac{\pi}{5}. \end{aligned}$$

■

Table A1. MISE values and average selection of N and M for spot covariance estimates obtained via (2) in the feasible and unfeasible cases.

ν_1, ν_2	Feasible		Unfeasible	
	N, M	MISE	N, M	MISE
1.5, 1	591, 3	$8.698 \cdot 10^{-4}$	580, 3	$8.238 \cdot 10^{-4}$
2.5, 1	540, 3	$8.895 \cdot 10^{-4}$	523, 3	$8.506 \cdot 10^{-4}$

Appendix C. Normality of the estimation error in Theorem 3.1

We study the quality of the Normal approximation for the standardized estimation error from Theorem 3.1 with finite samples, using simulations. Specifically, we employ 1000 daily trajectories generated under the baseline scenario of Section 4.1.2, which satisfies the Theorem's assumptions. We consider three different values of t , corresponding to 1, 2 and 3 hours, respectively. For each t considered, we evaluate the quality of the Gaussian approximation of the standardized estimation error, that is,

$$\mathcal{E}_{n,N,M}(t) := \rho(n)^{-1/4} M^{-1/2} \frac{\tilde{\Sigma}_{n,N,M}^{1,2}(t) - \Sigma^{1,2}(t)}{\sqrt{\Lambda(t)}},$$

where

$$\begin{aligned} \Lambda(t) := & \frac{1}{3c} \left(\Sigma^{1,1}(t) \Sigma^{2,2}(t) + \left(\Sigma^{1,2}(t) \right)^2 \right) + \frac{1}{a^2 c} \frac{T}{3} \gamma_{1,2}^2(t) \\ & + \frac{c}{9} \left(\Sigma^{1,1}(t) \frac{\rho(n_2)}{\rho(n)} \xi_2 + \Sigma^{2,2}(t) \frac{\rho(n_1)}{\rho(n)} \xi_1 \right) \\ & + \frac{c^3}{15} \frac{\rho(n_1) \rho(n_2)}{\rho^2(n)} \xi_1 \xi_2. \end{aligned}$$

To compute $\mathcal{E}_{n,N,M}$, we use tick-by-tick observations and set $(N, M) = (817, 3)$ when $\nu_1 = 1.5$ and $(N, M) = (740, 3)$ when $\nu_1 = 2.5$, based on the results in Table A1. Figures A1 and A2 contain quantile-quantile plots that show the quality of the Normal approximation of the standardized estimation error for the different values of t considered.

Appendix D. Feasible selection of the cutting frequencies for the spot covariance estimator

We evaluate the accuracy of the feasible method for selecting the pair (N, M) for the spot covariance estimator in a simulation study, conducted under the same data-generating process employed in Section 4.1.2. The constraint region R and the hyperparameters (ϵ, ϑ, K) are the same as in Section 4.1.2. Spot covariance estimates are obtained on the 30-minute grid.

Table A1 compares unfeasible optimal MISE values with those resulting from the application of the feasible method. The comparison suggests that the latter provides a very satisfactory performance, for both values of ν_1 . In particular, we observe that it leads to the same value of $M = 3$ as in the unfeasible case, while it slightly overestimates the optimal N , similarly to what happens when it is applied for estimating spot betas (see Table A1).

Appendix E. List of stocks employed in Section 5

- (1) **AABV** – AbbVie
- (2) **AAPL** – Apple

- (3) **ABT** – Abbott Laboratories
- (4) **ACN** – Accenture
- (5) **ADBE** – Adobe
- (6) **ADI** – Analog Devices
- (7) **AMAT** – Applied Materials
- (8) **AMD** – Advanced Micro Devices
- (9) **AMGN** – Amgen
- (10) **AMT** – American Tower Corporation
- (11) **AMZN** – Amazon.com
- (12) **ANET** – Arista Networks
- (13) **AVGO** – Broadcom
- (14) **AXP** – American Express Company
- (15) **BA** – The Boeing Company
- (16) **BAC** – Bank of America Corporation
- (17) **BLK** – BlackRock
- (18) **BMJ** – Bristol-Myers Squibb Company
- (19) **BRKB** – Berkshire Hathaway
- (20) **BSX** – Boston Scientific Corporation
- (21) **C** – Citigroup
- (22) **CAT** – Caterpillar
- (23) **CB** – Chubb Limited
- (24) **CDNS** – Cadence Design Systems
- (25) **CI** – The Cigna Group
- (26) **CMCSA** – Comcast Corporation
- (27) **CMG** – Chipotle Mexican Grill
- (28) **COP** – ConocoPhillips
- (29) **COST** – Costco Wholesale Corporation
- (30) **CRM** – Salesforce
- (31) **CSCO** – Cisco Systems
- (32) **CVX** – Chevron Corporation
- (33) **DE** – Deere & Company
- (34) **DHR** – Danaher Corporation
- (35) **DIS** – The Walt Disney Company
- (36) **DUK** – Duke Energy Corporation
- (37) **ETN** – Eaton Corporation
- (38) **FB** – Meta Platforms
- (39) **GE** – General Electric Company
- (40) **GILD** – Gilead Sciences
- (41) **GOOGL** – Alphabet
- (42) **GS** – The Goldman Sachs Group
- (43) **HD** – The Home Depot
- (44) **HON** – Honeywell International
- (45) **IBM** – International Business Machines Corporation
- (46) **INTC** – Intel Corporation
- (47) **ISRG** – Intuitive Surgical
- (48) **JNJ** – Johnson & Johnson
- (49) **JPM** – JPMorgan Chase & Co.
- (50) **KO** – The Coca-Cola Company
- (51) **LLY** – Eli Lilly and Company
- (52) **LMT** – Lockheed Martin Corporation
- (53) **LRCX** – Lam Research Corporation
- (54) **MA** – Mastercard Incorporated
- (55) **MCD** – McDonald's Corporation
- (56) **MDLZ** – Mondelez International
- (57) **MDT** – Medtronic
- (58) **MO** – Altria Group
- (59) **MRK** – Merck
- (60) **MS** – Morgan Stanley
- (61) **MSFT** – Microsoft Corporation
- (62) **MU** – Micron Technology
- (63) **NEE** – NextEra Energy
- (64) **NFLX** – Netflix
- (65) **NKE** – NIKE
- (66) **NOW** – ServiceNow
- (67) **NVDA** – NVIDIA Corporation
- (68) **ORCL** – Oracle Corporation
- (69) **PANW** – Palo Alto Networks
- (70) **PEP** – PepsiCo
- (71) **PFE** – Pfizer
- (72) **PG** – The Procter & Gamble Company
- (73) **PGR** – The Progressive Corporation

- (74) **PM** – Philip Morris International
- (75) **QCOM** – Qualcomm Incorporated
- (76) **REGN** – Regeneron Pharmaceuticals
- (77) **RTX** – Raytheon Technologies Corporation
- (78) **SBUX** – Starbucks Corporation
- (79) **SCHW** – The Charles Schwab Corporation
- (80) **SHW** – The Sherwin-Williams Company
- (81) **SNPS** – Synopsys
- (82) **SO** – The Southern Company
- (83) **SPGI** – S&P Global
- (84) **SYK** – Stryker Corporation
- (85) **TJX** – The TJX Companies
- (86) **TMO** – Thermo Fisher Scientific
- (87) **TMUS** – T-Mobile US
- (88) **TSLA** – Tesla
- (89) **TXN** – Texas Instruments Incorporated
- (90) **UBER** – Uber Technologies
- (91) **UNH** – UnitedHealth Group Incorporated
- (92) **UNP** – Union Pacific Corporation
- (93) **UPS** – United Parcel Service
- (94) **V** – Visa
- (95) **VRTX** – Vertex Pharmaceuticals Incorporated
- (96) **VZ** – Verizon Communications
- (97) **WFC** – Wells Fargo & Company
- (98) **WMT** – Walmart
- (99) **XOM** – Exxon Mobil Corporation
- (100) **ZTS** – Zoetis